Spectral–Spatial Reweighted Robust Nonlinear Unmixing for Hyperspectral Images Based on an Extended Multilinear Mixing Model

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***Abstract*— Recently, nonlinear unmixing algorithms have attracted special attention in hyperspectral image (HSI) processing. However, the inherent wavelength-dependent nonlinear intensity and noise effects in real HSIs are often overlooked, and the spatial information of HSIs has not been fully utilized in current studies. In this article, we propose a spectral–spatial reweighted robust nonlinear unmixing algorithm to solve the above problems. First, a robust unmixing method is built on an extended multilinear mixing model (EMLM), which employs the vectorized nonlinear parameters to describe the nonlinear intensity varying along with spectral bands and adopts the $\ell_{2,1}$-norm-based loss function to suppress the influence of noise. Second, to fully exploit the spectral–spatial information of HSIs, the nonlinear unmixing problem is reformulated with two regularizers. Specifically, a reweighted collaborative sparse regularizer is used to make the pixels in a superpixel-based neighborhood share the same subset of endmembers and have similar abundances because the neighboring pixels are usually composed of several materials in similar proportions, and a reweighted spectral total variation regularizer is utilized to improve the spectral–spatial smoothness of the vectorized nonlinear parameters by considering the local-region similarities of the nonlinear mixing effects. Finally, the constrained optimization problem is solved by the alternating direction method of multipliers (ADMM). The experimental results on simulated, semisimulated, and real hyperspectral datasets demonstrate that the proposed method outperforms several state-of-the-art nonlinear unmixing methods.**

***Index Terms*— Hyperspectral images (HSIs), nonlinear mixing models (NMMs), robust nonlinear unmixing, superpixel, total variation (TV).**

I. Introduction

HYPERSPECTRAL imaging spectrometers on aerospace platforms collect both spectral and spatial information of surface materials, which forms the 3-D data cubes (DCs) of hyperspectral images (HSIs). HSIs record not only the material spatial distributions, but also narrow spectral bands covering the typical visible and near-infrared wavelength internals [1]. However, due to the limited spatial resolution of HSIs, mixed pixels defined as the mixtures of pure material spectra (i.e., endmembers) in certain proportional coefficients (i.e., abundances) generally exist [2]. The main tasks of hyperspectral unmixing (HU) are composed of endmember extraction and abundance estimation.

The linear mixture model (LMM) assumes that the incoming light is only scattered by a material once before reaching the sensor, and thus, an observed pixel can be modeled as a linear combination of the endmembers. With straightforward mathematical representation and clear physical meaning, the LMM has been widely used for unmixing [3], but it is often considered to be only suitable for the simple checkerboard-like scenes. In complicated regions covered by the intimate mixtures of microscopic mineral particles and the vegetated regions of huge 3-D structures with the multilayer mixtures, the incoming light could interact with more than one material, leading to prominent nonlinear mixing effects [4]. Therefore, the nonlinear mixing models (NMMs) and corresponding methods are essential to tackle the unmixing problems properly in real scenarios [5].

To deal with the nonlinearity in spectral mixing, the NMMs-based unmixing methods have been emerging over the past decades. Radiative transfer models such as the Hapke model [6] associate the reflectance to a set of real-world physical parameters to perform nonlinear unmixing for the intimate mixtures. However, these models have obvious limitations in the acquisition of their inherent physical parameters for practical applications. To alleviate the over-reliance on physical parameters, the bilinear mixture models (BMMs) extend the LMM by introducing an additional bilinear term to represent the second-order scattering between materials, where the bilinear term is approximated as the Hadamard product between endmember vectors. The classic BMMs include the Nascimento model (NM) [7], the Fan model (FM) [8], the generalized bilinear model (GBM) [9], and the polynomial post-nonlinear mixing model (PPNM) [10]. Although the BMMs-based nonlinear unmixing algorithms have been extensively studied [11], [12], the higher order scatterings should be considered in some complex scenes, like urban areas [13].

Taking multiple scatterings into account, the multilinear mixing model (MLM) [14] includes all orders of interactions by introducing the scalar nonlinear parameters to describe the probability of light undergoing further interactions, and it can be seen as an extension of the PPNM. Based on

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the MLM, an unsupervised nonlinear unmixing method is proposed, termed MLMp [15], which estimates endmembers jointly with the abundances and nonlinear parameters. MLMp simplifies the objective function of the MLM to avoid the under-estimation of the nonlinear parameters and reduce the computational complexity. To mitigate the MLM-based unmixing methods falling into the local minima, a graph regularized MLM (G-MLM) [16] method imposes graph constraints on the abundances and nonlinear parameters, well exploiting the internal manifold structure of data. Although the abovementioned methods can account for the pixel-dependent nonlinear intensity and have achieved excellent unmixing performance, they are subject to a common implicit assumption that all the bands share the same mixing mode, which is contrary to the truth that the nonlinear contributions are wavelength-dependent [17], [18], [19].

In remote sensing, radiation is modeled as a function of wavelength, and the electromagnetic wave at each wavelength interacts with ground cover in different ways. Since the process of spectral mixing is closely related to the radiative transfer, multiple scatterings of different degrees may occur at the pixel's different bands simultaneously [18], [19]. To model the wavelength-dependent nonlinear intensity, a recently proposed extended multilinear mixing model (EMLM) [20] vectorizes the scalar nonlinear parameters of the MLM. A bandwise nonlinear spectral unmixing (BNLSU) method [20], based on the EMLM, performs better in explaining the nonlinear mixing effects at different bands and yields smaller data reconstruction errors (REs) than the other methods based on the traditional spectral mixture models. However, BNLSU is prone to overfitting and is sensitive to noise. In addition, BNLSU lacks the utilization of the rich spatial information of HSIs, which limits its unmixing accuracy.

Recently, some deep learning-based unmixing methods have emerged. Several representative works are briefly introduced here. In [21], the model-inspired neural networks for HU, based on the LMM and the iterative shrinkage thresholding algorithm, are discussed. By unrolling the optimization of the ℓ_p sparsity-constrained nonnegative matrix factorization (NMF) unmixing method, an LMM-based end-to-end deep neural network is introduced for HU [22]. Benefiting from the 3-D-convolutional neural network (CNN)-based networks jointly capturing the spectral–spatial information of HSIs, an unsupervised nonlinear spectral unmixing method based on a CNN autoencoder network is presented in [23]. In this article, the model-driven method is still used for unmixing due to its explicit physical interpretability.

Different unmixing methods, either the linear or nonlinear ones, suffer a lot from noise in the hyperspectral data. In particular, there are up to 20% corrupted channels of low signal-to-noise ratio (SNR) in real hyperspectral data, which severely affects the performance of the HU methods [24], [25]. To cope up with this issue, efforts have been dedicated to reducing the influence of noise. Compared with the commonly used ℓ_2-norm-based loss function, more robust metrics (such as the $\ell_{2,1}$-norm) have superior performances in dealing with highly noisy data [26], [27]. In the literature, different machine learning methods have achieved robustness to noise and outliers by using the $\ell_{2,1}$-norm [28], [29], [30], such as a recently proposed MLM-based robust nonlinear unmixing (R-MLM) algorithm [31]. However, R-MLM cannot model the wavelength-dependent nonlinear intensity, nor can it exploit the spatial information of HSIs.

On the other hand, the regularization of the unmixing optimization problem by resorting to HSIs' spectral–spatial information-based constraints is commonly considered to be a generic and effective strategy to mitigate overfitting and improve unmixing accuracy [32]. It is known that materials often appear in a local region of HSIs rather than being evenly distributed across the scene, which results in the sparsity of the abundances. Furthermore, the pixels usually exhibit strong spatial correlation, and the superiority of such prior information in improving the unmixing accuracy has been validated in the study of traditional linear unmixing [33], [34], [35]. However, the integration of the rich spectral and spatial information of HSIs into nonlinear unmixing remains a challenge. Direct use of additional spatial regularization may be sensitive to the parameter settings [36]. As a result, a popular means is to introduce weighting factors to the regularization terms in the optimization problems of unmixing instead [37], [38]. Of particular note is the spectral–spatial weighted sparse unmixing, which utilizes spectral and spatial information through two weighting factors [39].

Another concern is how to appropriately construct the pixel neighborhoods. A region-based structure preserving NMF is proposed in [40], where small homogeneous regions are generated with a graph cutting algorithm. Currently, some works are devoted to investigating superpixel-based spatial priors. To name a few, the group-structured prior information of HSIs is incorporated into the NMF optimization in [41], where the data are organized into spatial groups adaptively represented by superpixels. In [41], the superpixel-based local spatial information is integrated into the tensor factorization. A superpixel-based reweighted low-rank and total variation (TV) sparse unmixing algorithm is proposed in [43]. However, the above methods are based on the LMM, and they are limited in tackling the nonlinear unmixing problems.

Currently, few attempts have been made to integrate the spatial prior information of HSIs rationally into nonlinear unmixing. In [44], a kernel-based nonlinear unmixing model coupled with an ℓ_1-norm regularizer is proposed to promote the piecewise smoothness of abundance maps. Furthermore, a kernel-based nonlinear unmixing method in [45] exploits the spatial correlations within superpixels by employing a regional ℓ_1-norm regularizer. However, these methods ignore the local similarity of nonlinear intensities since the nonlinear mixing effects arising from the interactions between materials should be similar in local areas.

Most of the NMMs can only model the pixel-dependent nonlinear intensity, while the EMLM breaks this shackle and characterizes the wavelength-dependent nonlinear intensity by vectorizing the nonlinear parameters in the MLM. Constructed based on clear physical meanings, the EMLM provides a new idea to reflect the spectral mixing mechanism in real scenes [17], [18], [19], and it has a great potential for unmixing [20]. However, currently, there is little research work related to

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the EMLM, and there are some important flaws in their methods, prompting us to conduct more intensive research. In brief, two main problems of the existing unmixing methods are necessary to be deeply investigated and solved. First, the EMLM-based unmixing methods are prone to overfitting and lack robustness to noise. Second, it remains a great challenge to correctly express the local spatial similarity in nonlinear mixing and effectively integrate the spectral–spatial information of HSIs into nonlinear unmixing. To this end, the motivation of this article is to propose a nonlinear unmixing algorithm that can not only model the wavelength-dependent nonlinear intensity, but also overcome the noise effects and exploit the spectral–spatial information of HSIs.

To solve the above problems, we propose a spectral–spatial reweighted robust nonlinear unmixing algorithm, namely, S^2R^2NUA. First, to model the wavelength-dependent nonlinear intensity, the EMLM is employed to construct a nonlinear unmixing problem. Since a few bands in real hyperspectral data may suffer from severe noise, the $\ell_{2,1}$-norm is adopted to construct the loss function, which contributes to alleviating noise effects. Second, in accordance with the real-world situation that similar surface covers often occur in a local region, two regularizers are designed for the abundances and the vectorized nonlinear parameters, respectively. To be specific, the pixels in a superpixel-based neighborhood are considered to share the same subset of endmembers and have similar abundances, which is achieved by means of a reweighted collaborative sparse regularizer. Further, the spectral–spatial smoothness of the vectorized nonlinear parameters is improved with the help of a reweighted spectral TV regularizer. Finally, the initial nonlinear unmixing problem is reformulated with the two reweighted regularizers and further solved by the alternating direction method of multipliers (ADMM) [46].

The main contributions of this article can be concisely summarized as follows.

1) Considering the wavelength-dependent nonlinear intensity and noise effects, we present an EMLM-based robust unmixing method that adopts the $\ell_{2,1}$-norm-based loss function. Further, we integrate the spectral–spatial information of HSIs into the nonlinear unmixing problem by using the two regularizers, i.e., a reweighted collaborative sparse regularizer and a reweighted spectral TV regularizer, to overcome overfitting and improve unmixing accuracy.
2) ADMM is applied to solve the constrained optimization problem. The experimental results on the simulated, semisimulated, and real datasets show that the unmixing results with higher accuracy and stronger robustness can be achieved by the proposed method in comparison with other state-of-the-art ones.

The remainder of this article is organized as follows. Section II briefly introduces the traditional spectral mixture models and the EMLM. Next, the problem formulation and the detailed implementation of the optimization algorithm are presented in Section III. In Section IV, experiments on the simulated, semisimulated, and real datasets are conducted. Finally, Section V gives the conclusion of this article.

II. Mixture Models

In this section, we briefly review several widely used spectral mixture models and present the corresponding mathematical representations.

A. Traditional Spectral Mixture Models

Let $\boldsymbol{X} = [\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots, \boldsymbol{x}_N] \in \mathbb{R}^{L\times N}$ be the hyperspectral data matrix with L bands and N pixels that are constituted by M endmembers $\boldsymbol{E} = [\boldsymbol{e}_1, \boldsymbol{e}_2, \ldots, \boldsymbol{e}_M] \in \mathbb{R}^{L\times M}$, and the corresponding abundance matrix is $\boldsymbol{A} = [\boldsymbol{a}_1, \boldsymbol{a}_2, \ldots, \boldsymbol{a}_N] \in \mathbb{R}^{M\times N}$. The general mathematical expression of the LMM is

$$\boldsymbol{x}_j = \sum_{i=1}^{M} \boldsymbol{e}_i a_{i,j} + \boldsymbol{\varepsilon}_j \tag{1}$$

where $a_{i,j}$ is the ith endmember's abundance coefficient in $\boldsymbol{x}_j$, and $\boldsymbol{\varepsilon}_j \in \mathbb{R}^{L\times 1}$ is the additional noise. Two commonly used physical constraints are the abundance nonnegative constraint (ANC), i.e., $a_{i,j} \geq 0$, and the abundance sum-to-one constraint (ASC), i.e., $\sum_{i=1}^{M} a_{i,j} = 1$.

Accounting for only the secondary scatterings between materials, the BMMs are modeled as the LMM superimposed with an additional bilinear term approximated by the Hadamard product between endmember vectors. The ANC and ASC are also satisfied. A general mathematical representation of the BMMs can be given as

$$\boldsymbol{x}_j = \sum_{i=1}^{M} \boldsymbol{e}_i a_{i,j} + \sum_{i=1}^{t_1}\sum_{k=t_2}^{M} b_{i,k,j}(\boldsymbol{e}_i \odot \boldsymbol{e}_k)a_{i,j}a_{k,j} + \boldsymbol{\varepsilon}_j \tag{2}$$

where $b_{i,k,j}$ denotes the nonlinear parameter, whose definition depends on the specific mixture model. In the FM [8], $t_1 = M-1$, $t_2 = i+1$, and $b_{i,k,j} = 1$. By relaxing the FM's strict bilinear terms, the GBM [9] can be more flexible by using a variable $0 \leq b_{i,k,j} \leq 1$. Compared with the GBM, the PPNM [10] further considers the self-interaction between the same type of endmember, accordingly, $t_1 = M$, $t_2 = 1$, and $b_{i,k,j} \in \mathbb{R}$.

The MLM [14] further generalizes the PPNM to include all orders of interactions between endmembers, based on the following four assumptions: 1) the incoming light interacts with at least one material in the scene; 2) for the jth pixel, the probabilities of undergoing further interactions and reaching the sensor are p_j and $(1-p_j)$, respectively; 3) the probability of interacting with one material is proportional to its corresponding abundance; and 4) the intensity of the incoming light scattered by a material depends on the material's albedo approximated by the reflectance [14]. Hence, the formulation of the MLM model is written as

$$\begin{aligned}\boldsymbol{x}_j &= (1-p_j)\sum_{i=1}^{M} \boldsymbol{e}_i a_{i,j} + (1-p_j)p_j \sum_{i,k=1}^{M} (\boldsymbol{e}_i \odot \boldsymbol{e}_k)a_{i,j}a_{k,j} \\ &\quad + (1-p_j)p_j^2 \sum_{i,k,l=1}^{M} (\boldsymbol{e}_i \odot \boldsymbol{e}_k \odot \boldsymbol{e}_l)a_{i,j}a_{k,j}a_{l,j} + \cdots + \boldsymbol{\varepsilon}_j \\ &= (1-p_j)\boldsymbol{y}_j + p_j \boldsymbol{y}_j \odot \boldsymbol{x}_j + \boldsymbol{\varepsilon}_j \end{aligned} \tag{3}$$

where $\boldsymbol{y}_j = \sum_{i=1}^{M} \boldsymbol{e}_i a_{i,j} = \boldsymbol{E}\boldsymbol{a}_j$ computes the linear component, and the ANC and ASC are also imposed. In addition,

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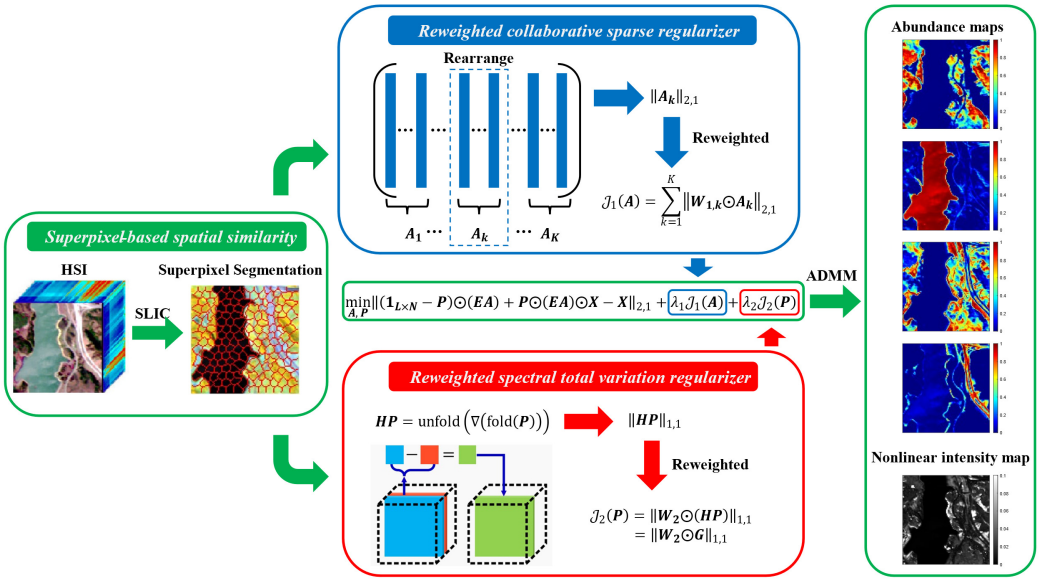


Fig. 1. Flowchart of S^{2}R^{2}NUA.

the scalar nonlinear parameter $p_j (j = 1, 2, \ldots, N)$ is less than 1 and varies along with pixels. It is noted that $p_j < 0$ is also allowed in (3), which increases the model's flexibility [14].

B. Extended MLM Model

To get rid of the drawback stemming from the assumption that all the bands share the same mixing mode, it is reasonable to vectorize the scalar nonlinear parameters of the MLM to model the wavelength-dependent nonlinear intensity. The EMLM [20] can be derived as

$$\boldsymbol{x}_j = \left(\mathbf{1}_L - \boldsymbol{P}_j\right) \odot \left(\boldsymbol{E}\boldsymbol{a}_j\right) + \boldsymbol{P}_j \odot \left(\boldsymbol{E}\boldsymbol{a}_j\right) \odot \boldsymbol{x}_j + \boldsymbol{\varepsilon}_j \quad (4)$$

where $\mathbf{1}_L \in \mathbb{R}^L$ denotes a vector of all 1s, and $\boldsymbol{P}_j \in \mathbb{R}^{L\times 1}$ denotes the vectorized nonlinear parameter for the jth pixel.

Referring to [15] and [20], the optimization problem based on (4) can be defined as

$$\min_{\boldsymbol{A},\boldsymbol{P}} \sum_{j=1}^{N} \left\| \left(\mathbf{1}_L - \boldsymbol{P}_j\right) \odot \left(\boldsymbol{E}\boldsymbol{a}_j\right) + \boldsymbol{P}_j \odot \left(\boldsymbol{E}\boldsymbol{a}_j\right) \odot \boldsymbol{x}_j - \boldsymbol{x}_j \right\|_2^2$$
$$\text{s.t. } a_{i,j} \geq 0, \quad \sum_{i=1}^{M} a_{i,j} = 1, \quad P_{l,j} < 1 \quad (5)$$

where the ANC and ASC are also satisfied and $P_{l,j} < 1$ measures the degree of multiple scattering at the lth band for the jth pixel.

III. Proposed Method

In this section, the proposed method is detailed in terms of the construction of the two regularizers and the derivation of the optimization problem via ADMM. The flowchart is shown in Fig. 1, taking a commonly used Jasper Ridge HSI as an example. The process of our method can be roughly divided into three parts, namely, first performing the superpixel segmentation of HSIs, then constructing two superpixel-based reweighted regularizers, and finally solving the constrained optimization problem by ADMM.

First, the simple linear iterative clustering (SLIC) [48] is applied to generate superpixels. Then, the abundance matrix $\boldsymbol{A}$ is rearranged according to the generated superpixel labels, i.e., $\boldsymbol{A} = [\boldsymbol{A}_1, \boldsymbol{A}_2, \ldots, \boldsymbol{A}_k, \ldots, \boldsymbol{A}_K]$, where $\boldsymbol{A}_k$ denotes an abundance submatrix corresponding to the pixels' abundances in the kth superpixel and K is the number of superpixels. A reweighted collaborative sparse regularizer $\mathcal{J}_1(\boldsymbol{A})$ is constructed by defining weight matrices $\boldsymbol{W}_{1,k} \in \mathbb{R}^{M\times N_k} (k = 1, 2, \ldots, K)$, where N_k denotes the number of pixels in the kth superpixel, to make the abundance submatrices have the collaborative sparseness and preserve the local spatial similarity.

Then, we impose a spectral TV regularizer on the nonlinear parameter matrix $\boldsymbol{P}$ to make it spectrally smooth, i.e., $\|\boldsymbol{H}\boldsymbol{P}\|_{1,1}$, where $\boldsymbol{H}$ is the linear difference operation on $\boldsymbol{P}$. Then, a new weight matrix $\boldsymbol{W}_2 \in \mathbb{R}^{L\times N}$ can be defined and employed to construct a reweighted spectral TV regularizer $\mathcal{J}_2(\boldsymbol{P})$ which functions in enhancing the spectral–spatial smoothness of $\boldsymbol{P}$.

Finally, the two regularizers are incorporated into the $\ell_{2,1}$-norm-based loss function, and the optimization problem is optimized via ADMM to generate the abundance maps and the nonlinear intensity map.

A. Problem Formulation

With the two regularizers of the abundances and vectorized nonlinear parameters, an S^{2}R^{2}NUA is proposed, where the EMLM-based data construction error is measured with the $\ell_{2,1}$-norm. In this sense, the corresponding optimization problem can be formulated as

$$\min_{\boldsymbol{A},\boldsymbol{P}} \|(\mathbf{1}_{L\times N} - \boldsymbol{P}) \odot (\boldsymbol{E}\boldsymbol{A}) + \boldsymbol{P} \odot (\boldsymbol{E}\boldsymbol{A}) \odot \boldsymbol{X} - \boldsymbol{X}\|_{2,1} + \lambda_1 \mathcal{J}_1(\boldsymbol{A}) + \lambda_2 \mathcal{J}_2(\boldsymbol{P})$$
$$\text{s.t. } a_{i,j} \geq 0, \quad \sum_{i=1}^{M} a_{i,j} = 1, \quad P_{l,j} < 1 \quad (6)$$

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where $\mathbf{1}_{L\times N} \in \mathbb{R}^{L\times N}$ denotes a matrix of all 1s, $\boldsymbol{P} = (\boldsymbol{P}_1, \boldsymbol{P}_2, \ldots, \boldsymbol{P}_j, \ldots, \boldsymbol{P}_N) \in \mathbb{R}^{L\times N}$ is the nonlinear parameter matrix, and λ_1 and λ_2 are the two parameters to balance the contribution of the two regularizers $\mathcal{J}_1(\boldsymbol{A})$ and $\mathcal{J}_2(\boldsymbol{P})$.

1) $\ell_{2,1}$-Norm-Based Loss Function: The ℓ_2-norm-based loss function in (7a) is commonly used to measure the similarity between the observed pixel and its reconstructed one, and $\boldsymbol{\mathcal{E}} = (\boldsymbol{\mathcal{E}}_1, \boldsymbol{\mathcal{E}}_2 \ldots, \boldsymbol{\mathcal{E}}_j, \ldots, \boldsymbol{\mathcal{E}}_N) \in \mathbb{R}^{L\times N}$ can be used to denote the data RE. However, in the case of the ℓ_2-norm-based loss function, the error subcomponent of the lth band being in the form of $((\sum_{j=1}^{N} \mathcal{E}_{l,j}^2)^{1/2})^2$ is easily dominated by a few bands with large errors [26], [29].

To overcome this drawback, we employ the $\ell_{2,1}$-norm-based loss function in (7b) to improve unmixing. In this case, the influence of noisy bands can be largely eliminated because $(\sum_{j=1}^{N} \mathcal{E}_{l,j}^2)^{1/2}$ is not squared. Equation (7b) can be considered to be equivalent to having an adaptive weight, i.e., $1/(\sum_{k=1}^{N} \mathcal{E}_{l,k}^2)^{1/2}$, based on (7a). A smaller weight can be automatically assigned to suppress the influence of the noisy bands [31], [47] if a larger error corresponding to a specific band is obtained

$$\sum_{j=1}^{N} \left\|\boldsymbol{\mathcal{E}}_j\right\|_2^2 = \sum_{l}^{L}\sum_{j=1}^{N} \mathcal{E}_{l,j}^2 \tag{7a}$$

$$\begin{aligned}\|\boldsymbol{\mathcal{E}}\|_{2,1} &= \sum_{l}^{L} \sqrt{\sum\nolimits_{j=1}^{N} \mathcal{E}_{l,j}^2} \\ &= \sum_{l}^{L}\sum_{j=1}^{N} \left(1 \Big/ \sqrt{\sum\nolimits_{k=1}^{N} \mathcal{E}_{l,k}^2}\right) \mathcal{E}_{l,j}^2.\end{aligned} \tag{7b}$$

2) Reweighted Collaborative Sparse Regularizer: Usually, local regions of HSIs can be segmented to discriminate neighboring pixels. Instead of using a window with a predefined size, we use the SLIC to generate an adaptive local spatial neighborhood for each pixel to better describe the spatial correlation. Then, the abundance matrix $\boldsymbol{A}$ is rearranged according to the generated superpixel labels, i.e., $\boldsymbol{A} = [\boldsymbol{A}_1, \boldsymbol{A}_2, \ldots, \boldsymbol{A}_k, \ldots, \boldsymbol{A}_K]$, where $\boldsymbol{A}_k$ denotes an abundance submatrix corresponding to the pixels' abundances in the kth superpixel and K is the number of superpixels.

Usually, the surface cover in a local region is composed of several materials in similar proportions. Therefore, the pixels in a region of the same superpixel may share the same subset of endmembers and have similar abundances. In our method, the collaborative sparsity of the abundances is enforced with the $\ell_{2,1}$-norm, and the reweighted strategy [49] is adopted to impose the similarity on the abundances. We can define weight matrices $\boldsymbol{W}_{1,k} \in \mathbb{R}^{M\times N_k} (k = 1, 2, \ldots, K)$ to integrate the superpixel-based spatial similarity into the EMLM-based unmixing framework

$$\left(\boldsymbol{W}_{1,k}^{(t+1)}\right)_{i,j} = \frac{1}{\left|f\left(a_{i,j}^{(t)}\right)\right| + \delta} \tag{8}$$

where $(\boldsymbol{W}_{1,k}^{(t+1)})_{i,j}$ is an element in the ith row and the jth column of $\boldsymbol{W}_{1,k}$ at the $(t+1)$th iteration, and δ is a small positive value that ensures that the denominator is nonzero. We can define the neighborhood $\Omega(j)$ as the eight pixels most similar to the jth pixel in the same superpixel. A function $f(\cdot)$ is used to normalize the abundances of the jth pixel through a weighted average approach and can be written as

$$f\left(a_{i,j}^{(t)}\right) = \frac{\sum_{q\in\Omega(j)} d_{j,q} a_{i,q}^{(t)}}{\sum_{q\in\Omega(j)} d_{j,q}} \tag{9}$$

where $d_{j,q}$ measures the similarity between the reference pixel $\boldsymbol{x}_j$ and one of the nonreference pixels in $\Omega(j)$ (i.e., $\boldsymbol{x}_q$). It can be calculated by combining the spectral angle distance (SAD) in (10) and spatial Euclidean distance in (11)

$$d_{\text{spe}} = \cos^{-1}\left(\frac{\boldsymbol{x}_j^{\top}\boldsymbol{x}_q}{\|\boldsymbol{x}_j\|_2\|\boldsymbol{x}_q\|_2}\right) \tag{10}$$

$$d_{\text{spa}} = \sqrt{\left(m_j - m_q\right)^2 + \left(n_j - n_q\right)^2} \tag{11}$$

where (m_j, n_j) and (m_q, n_q) denote the spatial coordinates of pixels, respectively.

And, $d_{j,q}$ can be formulated as

$$d_{j,q} = \exp\left(-\sqrt{d_{\text{spe}}^2 + \left(d_{\text{spa}}/\kappa\right)^2 \kappa_{\text{s}}^2}\right) \tag{12}$$

where κ and κ_{s} are the two parameters that control the average size of the superpixels and weight the relative importance between the spectral and spatial similarity, respectively. Finally, the reweighted collaborative sparse regularizer of the abundances can be derived as

$$\mathcal{J}_1(\boldsymbol{A}) = \sum_{k=1}^{K} \left\|\boldsymbol{W}_{1,k} \odot \boldsymbol{A}_k\right\|_{2,1}. \tag{13}$$

3) Reweighted Spectral TV Regularizer: Although the nonlinear intensity in the same pixel varies along with each band, the adjacent bands tend to have similar nonlinear intensities [51]. The rationality of this assumption has been discussed in the previous literatures [17], [18], [19]. In addition, the nonlinear intensities at the same band are similar among locally neighboring pixels. Therefore, it is reasonable that the vectorized nonlinear parameters have the spectral–spatial smoothness.

In detail, the matrix $\boldsymbol{P} \in \mathbb{R}^{L\times N}$ is folded along the spatial mode to $\mathcal{P} = \text{fold}(\boldsymbol{P}) \in \mathbb{R}^{L\times h\times w}$, where h and w are the spatial dimensions of HSIs, i.e., height and width, respectively. Let ∇ denote the difference operation calculated along the spectral mode on $\mathcal{P}$, and then, $\mathcal{G} = \nabla\mathcal{P} \in \mathbb{R}^{L\times h\times w}$ is the gradient tensor. $\boldsymbol{G} = \text{unfold}(\mathcal{G}) \in \mathbb{R}^{L\times N}$ in the following equation represents the unfolding matrices of these gradient maps along the spectral mode [52]:

$$\boldsymbol{H}\boldsymbol{P} = \text{unfold}(\nabla(\text{fold}(\boldsymbol{P}))) = \boldsymbol{G} \tag{14}$$

where $\boldsymbol{H}$ is the linear difference operation on $\boldsymbol{P}$.

Benefiting from the sparsity-induced property of the ℓ_1-norm, the spectral TV regularizer $\|\boldsymbol{G}\|_{1,1} = \sum_{j=1}^{N} \|\boldsymbol{G}_j\|_1$ enforces the spectral smoothness of the vectorized nonlinear parameters, where $\boldsymbol{G}_j \in \mathbb{R}^L$ represents the gradient vector for the jth pixel. Moreover, due to the spatial correlation being retained by the linear operation $\boldsymbol{H}$, the gradient matrix $\boldsymbol{G}$ still

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contains spatial information. At the $(t+1)$th iteration, a new weight matrix $\boldsymbol{W}_2 \in \mathbb{R}^{L\times N}$ can be updated like (8)

$$\left(\boldsymbol{W}_2^{(t+1)}\right)_{l,j} = \frac{1}{\left|f\left(G_{l,j}^{(t)}\right)\right| + \delta} \tag{15}$$

where $(\boldsymbol{W}_2^{(t+1)})_{l,j}$ is an element in the lth row and the jth column of $\boldsymbol{W}_2$. Finally, a reweighted spectral TV regularizer of the vectorized nonlinear parameters $\mathcal{J}_2(\boldsymbol{P})$ is defined as

$$\mathcal{J}_2(\boldsymbol{P}) = \|\boldsymbol{W}_2 \odot \ \boldsymbol{G}\|_{1,1} = \|\boldsymbol{W}_2 \odot (\boldsymbol{H}\boldsymbol{P})\|_{1,1}. \tag{16}$$

B. Optimization Using ADMM

By using the reweighting strategy, the spectral–spatial information of HSIs in nonlinear unmixing can be utilized in the proposed S^{2}R^{2}NUA. The constrained optimization problem in (6) for nonlinear unmixing is constructed by adopting the two regularizers in (13) and (16). In this article, we iteratively optimize (6) in an outer–inner looping scheme [39]. Specifically, the inner loop updates $\boldsymbol{A}$ and $\boldsymbol{P}$ via ADMM with the two fixed weights $\boldsymbol{W}_1$ and $\boldsymbol{W}_2$ that are computed in the outer loop. After the primal variables are decomposed into several new variables corresponding to the items of the optimization problem, ADMM updates all variables alternately. Optimal solutions can be effectively searched by ADMM after the original complicated optimization problem is transformed into a set of simple subproblems. After introducing auxiliary variables $\boldsymbol{V}_1, \ldots, \boldsymbol{V}_5$, we can redefine (6) as

$$\begin{aligned}
&\arg\min_{\boldsymbol{A},\boldsymbol{P},\boldsymbol{V}_1,\ldots,\boldsymbol{V}_5} \|\boldsymbol{V}_1\|_{2,1} + \lambda_1 \sum_{k=1}^{K} \|\boldsymbol{W}_{1,k} \odot \boldsymbol{V}_{2,k}\|_{2,1} \\
&\qquad + \lambda_2 \|\boldsymbol{W}_2 \odot \boldsymbol{V}_4\|_{1,1} \\
&\qquad + \iota_{\{\mathbf{1}_N^\top\}}\left(\mathbf{1}_M^\top \boldsymbol{A}\right) + \iota_{\mathbb{R}_+^{M\times N}}(\boldsymbol{V}_2) + \iota_{\{\boldsymbol{V}_5 | \boldsymbol{V}_5 < \mathbf{1}_{L\times N}\}}(\boldsymbol{V}_5) \\
&\text{s.t. } \boldsymbol{V}_1 = (\mathbf{1}_{L\times N} - \boldsymbol{P}) \odot (\boldsymbol{E}\boldsymbol{A}) + \boldsymbol{P} \odot (\boldsymbol{E}\boldsymbol{A}) \odot \boldsymbol{X} - \boldsymbol{X} \\
&\qquad \boldsymbol{V}_2 = \boldsymbol{A}, \quad \boldsymbol{V}_3 = \boldsymbol{P}, \ \boldsymbol{V}_4 = \boldsymbol{H}\boldsymbol{V}_3, \quad \boldsymbol{V}_5 = \boldsymbol{P}
\end{aligned} \tag{17}$$

where $\iota_{\{\mathbf{1}_N^\top\}}(\mathbf{1}_M^\top A)$, $\iota_{\mathbb{R}_+^{M\times N}}(A)$, and $\iota_{\{\boldsymbol{P}|\boldsymbol{P}<\mathbf{1}_{L\times N}\}}(\boldsymbol{P})$ are the three indicator functions that impose the ASC and ANC on the abundances and bound the nonlinear parameters less than 1, respectively. The augmented Lagrangian function of (17) can be written as

$$\begin{aligned}
&\mathcal{L}(\boldsymbol{A}, \boldsymbol{P}, \boldsymbol{V}_1, \ldots, \boldsymbol{V}_5, \boldsymbol{U}_1, \ldots, \boldsymbol{U}_5) \\
&= \|\boldsymbol{V}_1\|_{2,1} + \lambda_1 \sum_{k=1}^{K} \|\boldsymbol{W}_{1,k} \odot \boldsymbol{V}_{2,k}\|_{2,1} + \lambda_2 \|\boldsymbol{W}_2 \odot \boldsymbol{V}_4\|_{1,1} \\
&\quad + \iota_{\{\mathbf{1}_N^\top\}}\left(\mathbf{1}_M^\top \boldsymbol{A}\right) + \iota_{\mathbb{R}_+^{M\times N}}(\boldsymbol{V}_2) + \iota_{\{\boldsymbol{V}_5 | \boldsymbol{V}_5 < \mathbf{1}_{L\times N}\}}(\boldsymbol{V}_5) \\
&\quad + \frac{\mu}{2}\|(\mathbf{1}_{L\times N} - \boldsymbol{P}) \odot (\boldsymbol{E}\boldsymbol{A}) \\
&\qquad + \boldsymbol{P} \odot (\boldsymbol{E}\boldsymbol{A}) \odot \boldsymbol{X} - \boldsymbol{X} - \boldsymbol{V}_1 - \boldsymbol{U}_1\|_F^2 \\
&\quad + \frac{\mu}{2}\|\boldsymbol{A} - \boldsymbol{V}_2 - \boldsymbol{U}_2\|_F^2 + \frac{\mu}{2}\|\boldsymbol{P} - \boldsymbol{V}_3 - \boldsymbol{U}_3\|_F^2 \\
&\quad + \frac{\mu}{2}\|\boldsymbol{H}\boldsymbol{V}_3 - \boldsymbol{V}_4 - \boldsymbol{U}_4\|_F^2 + \frac{\mu}{2}\|\boldsymbol{P} - \boldsymbol{V}_5 - \boldsymbol{U}_5\|_F^2
\end{aligned} \tag{18}$$

where $\boldsymbol{U}_1, \ldots, \boldsymbol{U}_5$ are the scaled dual variables, and μ is a penalty parameter with a small positive value controlling the convergence speed of the algorithm. ADMM minimizes the above problem by separating it into simple optimization of several subproblems with respect to each unknown variable. We alternately minimize one variable with the others fixed and give the sequential optimization for variables $\boldsymbol{A}$, $\boldsymbol{P}$, $\boldsymbol{V}_1, \ldots, \boldsymbol{V}_5$ and scaled dual variables $\boldsymbol{U}_1, \ldots, \boldsymbol{U}_5$.

1) Minimization With Respect to $\boldsymbol{A}$: At the $(t+1)$th iteration, discarding the terms irrelevant with $\boldsymbol{A}$, we update $\boldsymbol{A}^{(t+1)}$ by minimizing the following optimization subproblem:

$$\begin{aligned}
\boldsymbol{A}^{(t+1)} \leftarrow \arg\min_{\boldsymbol{A}} &\frac{\mu}{2}\left\|\boldsymbol{A} - \boldsymbol{V}_2^{(t)} - \boldsymbol{U}_2^{(t)}\right\|_F^2 + \iota_{\{\mathbf{1}_N^\top\}}\left(\mathbf{1}_M^\top \boldsymbol{A}\right) \\
&+ \frac{\mu}{2}\left\|\left(\mathbf{1}_{L\times N} - \boldsymbol{P}^{(t)}\right) \odot (\boldsymbol{E}\boldsymbol{A}) + \boldsymbol{P}^{(t)}\right. \\
&\left. \odot (\boldsymbol{E}\boldsymbol{A}) \odot \boldsymbol{X} - \boldsymbol{X} - \boldsymbol{V}_1^{(t)} - \boldsymbol{U}_1^{(t)}\right\|_F^2.
\end{aligned} \tag{19}$$

Update the optimization problem pixel-wisely as

$$\begin{aligned}
\boldsymbol{a}_j^{(t+1)} \leftarrow \arg\min_{\boldsymbol{a}_j} &\iota_{\{1\}}\left(\mathbf{1}_M^\top \boldsymbol{a}_j\right) + \frac{\mu}{2}\left\|\boldsymbol{E}^{(t)}\boldsymbol{a}_j - \boldsymbol{x}_j^{(t)}\right\|_2^2 \\
&+ \frac{\mu}{2}\left\|\boldsymbol{a}_j - (\boldsymbol{V}_2)_j^{(t)} - (\boldsymbol{U}_2)_j^{(t)}\right\|_2^2
\end{aligned} \tag{20}$$

where $\boldsymbol{E}^{(t)} = \boldsymbol{E} \odot ((\mathbf{1}_L - \boldsymbol{P}_j^{(t)} + \boldsymbol{P}_j^{(t)} \odot \boldsymbol{x}_j)\mathbf{1}_M^\top)$ is a pixelwise matrix depending on $\boldsymbol{P}_j^{(t)}$, and $\boldsymbol{x}_j^{(t)} = \boldsymbol{x}_j + (\boldsymbol{V}_1)_j^{(t)} + (\boldsymbol{U}_1)_j^{(t)}$ where $(\boldsymbol{V}_i)_j^{(t)}$ and $(\boldsymbol{U}_i)_j^{(t)}$ denote the columns of $\boldsymbol{V}_i$ and $\boldsymbol{U}_i$ $(i = 1, 2)$, respectively.

According to [33], the solution is derived by solving a quadratic problem with equality constraints, given by

$$\boldsymbol{a}_j^{(t+1)} \leftarrow \ \boldsymbol{B}^{-1}\boldsymbol{\omega} - \boldsymbol{C}\left(\mathbf{1}_M^\top \boldsymbol{B}^{-1}\boldsymbol{\omega} - 1\right) \tag{21}$$

where $\boldsymbol{B} = (\boldsymbol{E}^{(t)})^\top \boldsymbol{E}^{(t)} + \ \boldsymbol{I}$, $\boldsymbol{C} = \boldsymbol{B}^{-1}\mathbf{1}_M(\mathbf{1}_M^\top \boldsymbol{B}^{-1}\mathbf{1}_M)^{-1}$, $\boldsymbol{\omega} = (\boldsymbol{E}^{(t)})^\top \boldsymbol{x}_j^{(t)} + (\boldsymbol{V}_2)_j^{(t)} - (\boldsymbol{U}_2)_j^{(t)}$, and $\boldsymbol{I}$ is an identity matrix.

2) Minimization With Respect to $\boldsymbol{P}$: The optimization of the subproblem with respect to $\boldsymbol{P}$ is given as

$$\begin{aligned}
\boldsymbol{P}^{(t+1)} \leftarrow \arg\min_{\boldsymbol{P}} &\frac{\mu}{2}\left\|\boldsymbol{P} - \boldsymbol{V}_3^{(t)} - \boldsymbol{U}_3^{(t)}\right\|_F^2 \\
&+ \frac{\mu}{2}\left\|\boldsymbol{P} - \boldsymbol{V}_5^{(t)} - \boldsymbol{U}_5^{(t)}\right\|_F^2 \\
&+ \frac{\mu}{2}\left\|\boldsymbol{Y}^{(t+1)} \odot \left(\mathbf{1}_{L\times N} - \boldsymbol{P} + \boldsymbol{Y}^{(t+1)} \odot \boldsymbol{X}\right)\right. \\
&\left. - \boldsymbol{X} - \boldsymbol{V}_1^{(t)} - \boldsymbol{U}_1^{(t)}\right\|_F^2
\end{aligned} \tag{22}$$

and the solution is given as

$$\boldsymbol{P}^{(t+1)} \leftarrow \frac{\left(\boldsymbol{Y}^{(t+1)} - \boldsymbol{Y}^{(t+1)} \odot \boldsymbol{X}\right) \odot \left(\boldsymbol{Y}^{(t+1)} - \boldsymbol{X}^{(t)}\right) + \boldsymbol{\alpha} + \boldsymbol{\beta}}{\left(\boldsymbol{Y}^{(t+1)} - \boldsymbol{Y}^{(t+1)} \odot \boldsymbol{X}\right)^2 + 2\mathbf{1}_{L\times N}} \tag{23}$$

where $\boldsymbol{Y}^{(t+1)} = \boldsymbol{E}\boldsymbol{A}^{(t+1)}$ is the linear mixture component, and $\boldsymbol{X}^{(t)} = \boldsymbol{X} + \boldsymbol{V}_1^{(t)} + \boldsymbol{U}_1^{(t)}$, $\boldsymbol{\alpha} = \ \boldsymbol{V}_3^{(t)} + \boldsymbol{U}_3^{(t)}$, and $\boldsymbol{\beta} = \boldsymbol{V}_5^{(t)} + \boldsymbol{U}_5^{(t)}$.

3) Minimization With Respect to $\boldsymbol{V}_1$:

$$\begin{aligned}
\boldsymbol{V}_1^{(t+1)} \leftarrow \arg\min_{\boldsymbol{V}_1} &\|\boldsymbol{V}_1\|_{2,1} \\
&+ \frac{\mu}{2}\left\|\boldsymbol{Y}^{(t+1)} \odot \left(\mathbf{1}_{L\times N} - \boldsymbol{P}^{(t+1)} + \boldsymbol{P}^{(t+1)} \odot \boldsymbol{X}\right)\right. \\
&\left. - \boldsymbol{X} - \boldsymbol{V}_1 - \boldsymbol{U}_1^{(t)}\right\|_F^2
\end{aligned} \tag{24}$$

whose solution is

$$\boldsymbol{V}_1^{(t+1)}(m, :) \leftarrow \text{vect-soft}(\boldsymbol{\gamma}(m, :), 1/\mu) \tag{25}$$

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where $\boldsymbol{\gamma} = \boldsymbol{Y}^{(t+1)} \odot (\boldsymbol{1}_{L\times N} - \boldsymbol{P}^{(t+1)} + \boldsymbol{P}^{(t+1)} \odot \boldsymbol{X}) - \boldsymbol{X} - \boldsymbol{V}_1 - \boldsymbol{U}_1^{(t)}$, and vect-soft$(\boldsymbol{u}, t)$ is the rowwise application of the vect-soft threshold function [26], i.e.,

$$\text{vect-soft}(\boldsymbol{u}, t) \equiv \frac{\boldsymbol{u}(\max\{\|\boldsymbol{u}\|_2 - t, 0\})}{\max\{\|\boldsymbol{u}\|_2 - t, 0\} + t}. \tag{26}$$

4) Minimization With Respect to $\boldsymbol{V}_2$: We update $\boldsymbol{V}_2^{(t+1)}$ in a superpixel-by-superpixel manner, for $k = 1, 2, \ldots, K$

$$\boldsymbol{V}_{2,k}^{(t+1)} \leftarrow \arg\min_{\boldsymbol{V}_{2,k}} \lambda_1 \left\| \boldsymbol{W}_{1,k}^{(t+1)} \odot \boldsymbol{V}_{2,k} \right\|_{2,1} + \iota_{\mathbb{R}_+^{M\times N}}(\boldsymbol{V}_{2,k}) + \frac{\mu}{2} \left\| \boldsymbol{A}^{(t+1)} - \boldsymbol{V}_{2,k} - \boldsymbol{U}_{2,k}^{(t)} \right\|_F^2. \tag{27}$$

The soft threshold solution of it is given as

$$\boldsymbol{V}_{2,k}^{(t+1)}(m,:) \leftarrow \text{vect-soft}\left(\left(\boldsymbol{A}^{(t+1)} - \boldsymbol{U}_{2,k}^{(t)} \right)(m,:), \frac{\lambda_1}{\mu} \boldsymbol{W}_{1,k}^{(t+1)}(m,n) \right). \tag{28}$$

Considering ANC, the final solution is written as

$$\boldsymbol{V}_{2,k}^{(t+1)} \leftarrow \max\left(\boldsymbol{V}_{2,k}^{(t+1)}, 0 \right). \tag{29}$$

5) Minimization With Respect to $\boldsymbol{V}_3$:

$$\boldsymbol{V}_3^{(t+1)} \leftarrow \arg\min_{\boldsymbol{V}_3} \frac{\mu}{2} \left\| \boldsymbol{P}^{(t+1)} - \boldsymbol{V}_3 - \boldsymbol{U}_3^{(t)} \right\|_F^2 + \frac{\mu}{2} \left\| \boldsymbol{H}\boldsymbol{V}_3 - \boldsymbol{V}_4^{(t)} + \boldsymbol{U}_4^{(t)} \right\|_F^2 \tag{30}$$

and (30) has the solution

$$\boldsymbol{V}_3^{(t+1)} \leftarrow \left(\boldsymbol{H}^\top \boldsymbol{H} + \boldsymbol{I} \right)^{-1} \left(\boldsymbol{P}^{(t+1)} - \boldsymbol{U}_3^{(t)} + \boldsymbol{H}^\top \boldsymbol{\varphi} \right) \tag{31}$$

where $\boldsymbol{\varphi} = \boldsymbol{V}_4^{(t)} + \boldsymbol{U}_4^{(t)}$. The fast Fourier transform is readily employed to solve it [52], [53].

6) Minimization With Respect to $\boldsymbol{V}_4$:

$$\boldsymbol{V}_4^{(t+1)} \leftarrow \arg\min_{\boldsymbol{V}_4} \lambda_2 \left\| \boldsymbol{W}_2^{(t+1)} \odot \boldsymbol{V}_4 \right\|_{1,1} + \frac{\mu}{2} \left\| \boldsymbol{H}\boldsymbol{V}_3^{(t+1)} - \boldsymbol{V}_4 - \boldsymbol{U}_4^{(t)} \right\|_F^2 \tag{32}$$

and the solution of (32) is given as

$$\boldsymbol{V}_4^{(t+1)} \leftarrow \text{soft}\left(\boldsymbol{H}\boldsymbol{V}_3^{(t+1)} - \boldsymbol{U}_4^{(t)}, \frac{\lambda_2}{\mu} \boldsymbol{W}_2^{(t+1)} \right) \tag{33}$$

where soft$(\boldsymbol{u}, \boldsymbol{a}) \equiv \text{sign}(\boldsymbol{u}) \max\{|\boldsymbol{u}| - \boldsymbol{a}, 0\}$ denotes the soft threshold function [33].

7) Minimization With Respect to $\boldsymbol{V}_5$:

$$\boldsymbol{V}_5^{(t+1)} \leftarrow \arg\min_{\boldsymbol{V}_5} \iota_{\{\boldsymbol{V}_5 | \boldsymbol{V}_5 < \boldsymbol{1}_{L\times N}\}}(\boldsymbol{V}_5) + \frac{\mu}{2} \left\| \boldsymbol{P}^{(t+1)} - \boldsymbol{V}_5 - \boldsymbol{U}_5^{(t)} \right\|_F^2 \tag{34}$$

and it has the solution

$$\boldsymbol{V}_5^{(t+1)} \leftarrow \min\left(\boldsymbol{P}^{(t+1)} - \boldsymbol{U}_5^{(t)}, 1 \right). \tag{35}$$

8) Minimization With Respect to $\boldsymbol{U}_1, \boldsymbol{U}_2, \ldots, \boldsymbol{U}_5$:

$$\boldsymbol{U}_1^{(t+1)} \leftarrow \boldsymbol{U}_1 + \boldsymbol{V}_1^{(t+1)} - \left(\boldsymbol{Y}^{(t+1)} \odot \left(\boldsymbol{1}_{L\times N} - \boldsymbol{P}^{(t+1)} + \boldsymbol{P}^{(t+1)} \odot \boldsymbol{X} \right) - \boldsymbol{X} \right) \tag{36}$$

$$\boldsymbol{U}_2^{(t+1)} \leftarrow \boldsymbol{U}_2 + \boldsymbol{V}_2^{(t+1)} - \boldsymbol{A}^{(t+1)} \tag{37}$$

$$\boldsymbol{U}_3^{(t+1)} \leftarrow \boldsymbol{U}_3 + \boldsymbol{V}_3^{(t+1)} - \boldsymbol{P}^{(t+1)} \tag{38}$$

$$\boldsymbol{U}_4^{(t+1)} \leftarrow \boldsymbol{U}_4 + \boldsymbol{V}_4^{(t+1)} - \boldsymbol{H}\boldsymbol{V}_3^{(t+1)} \tag{39}$$

$$\boldsymbol{U}_5^{(t+1)} \leftarrow \boldsymbol{U}_5^{(t)} + \boldsymbol{V}_5^{(t+1)} - \boldsymbol{P}^{(t+1)}. \tag{40}$$

The pseudo-code of ADMM for S^{2}R^{2}NUA is given in **Algorithm 1**.

Algorithm 1 Spectral-Spatial Reweighted Robust Nonlinear Unmixing Algorithm (S^{2}R^{2}NUA)

Input: Hyperspectral data $\boldsymbol{X} \in \mathbb{R}^{L\times N}$, endmember matrix $\boldsymbol{E} \in \mathbb{R}^{L\times M}$, and parameters κ and κ_s.
Output: Abundance matrix $\boldsymbol{A} \in \mathbb{R}^{M\times N}$ and nonlinear parameter matrix $\boldsymbol{P} \in \mathbb{R}^{L\times N}$.
Segmentation: $\boldsymbol{X} = [\boldsymbol{X}_1, \boldsymbol{X}_1, \ldots, \boldsymbol{X}_k, \ldots, \boldsymbol{X}_K]$ denotes K superpixels generated by SLIC.
Set $t, m = 0$, $\mu > 0$, $\lambda_1 > 0$, and $\lambda_2 > 0$.
Initialization: $\boldsymbol{A}^{(0)}$, $\boldsymbol{P}^{(0)}$, and $\boldsymbol{V}_n^{(0)}, \boldsymbol{U}_n^{(0)}$, $n = 1, \ldots, 5$.
While stopping conditions are not met, **do**
  $\left(\boldsymbol{W}_1^{(m)}\right)_{i,j} \leftarrow 1/\left(\left| f\left((\boldsymbol{A} - \boldsymbol{U}_2)_{i,j}^{(m)} \right) \right| + \delta\right)$,
  $\left(\boldsymbol{W}_2^{(m)}\right)_{i,j} \leftarrow 1/\left(\left| f\left((\boldsymbol{H}\boldsymbol{V}_3 - \boldsymbol{U}_4)_{i,j}^{(m)} \right) \right| + \delta\right)$.
  While stopping conditions are not met, **do**
    Update $\boldsymbol{A}^{(t+1)}$ and $\boldsymbol{P}^{(t+1)}$ with (21) and (23), respectively.
    Update $\boldsymbol{V}_n^{(t+1)}, n = 1, \ldots, 5$ with (25), (29), (31), (33), and (35), respectively.
    Update $\boldsymbol{U}_n^{(t+1)}, n = 1, \ldots, 5$ with (36)-(40), respectively. $t \leftarrow t + 1$.
  End
  $\boldsymbol{A}^{(m+1)} \leftarrow \boldsymbol{A}^{(t+1)}$, $\boldsymbol{U}_2^{(m+1)} \leftarrow \boldsymbol{U}_2^{(t+1)}$, $\boldsymbol{V}_3^{(m+1)} \leftarrow \boldsymbol{V}_3^{(t+1)}$, $\boldsymbol{U}_4^{(m+1)} \leftarrow \boldsymbol{U}_4^{(t+1)}$
  $m \leftarrow m + 1$.
End

C. Implementation Details

The implementation details of S^{2}R^{2}NUA are presented in this section. The vertex component analysis (VCA) [54] and the fully constrained least squares (FCLS) [55] methods can be applied to extract the endmembers and obtain the initialization $\boldsymbol{A}^{(0)}$ satisfying the ANC and ASC, respectively. Then, $\boldsymbol{A}^{(0)}$ is used to initialize $\boldsymbol{P}^{(0)}$ based on (5), and the bounded constraint $\boldsymbol{P}^{(0)} < \boldsymbol{1}_{L\times N}$ is imposed as well. Set the value of $\boldsymbol{V}_1^{(0)}, \ldots, \boldsymbol{V}_5^{(0)}$ according to (17) and initialize $\boldsymbol{U}_1^{(0)}, \ldots, \boldsymbol{U}_5^{(0)}$ with zero matrices of different sizes.

The stopping conditions include twofold: 1) the number of iterations for both inner and outer loops should not exceed the corresponding maximum number of iterations and 2) in each execution of the outer loop, the accordingly calculated residual errors *res* from the inner loop should not exceed the

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tolerance. According to [36], we define *res* as

$$res = \Big\| (\boldsymbol{EA}) \odot (\boldsymbol{1}_{L\times N} - \boldsymbol{P} + \boldsymbol{P} \odot \boldsymbol{X}) - \boldsymbol{X} + \boldsymbol{A} + \boldsymbol{HV}_3 + 2\boldsymbol{IP} - \sum_{i=1}^{5} \boldsymbol{V}_i \Big\|_{\mathrm{F}}. \quad (41)$$

Before conducting experiments, we give a brief analysis about the complexity of the proposed method. In S^{2}R^{2}NUA, two steps of updating $\boldsymbol{A}$ and $\boldsymbol{V}_3$ take the highest costs, and at each iteration, the computational complexities of them are $\mathcal{O}(MNL^2)$ and $\mathcal{O}(NL^2\log(L))$, respectively.

IV. Experimental Results

In this section, we conduct experiments on three types of hyperspectral datasets, i.e., the simulated, semisimulated, and real datasets, to validate the superiority of the proposed method over other state-of-the-art ones.

A. Compared Methods and Performance Metrics

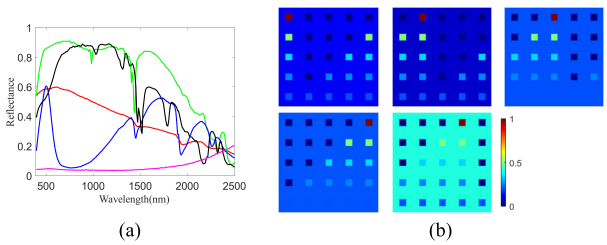
Since the proposed method is constructed based on explicit physical meanings, it makes sense to select the relevant physics-driven unmixing algorithms for comparative experiments. In this article, S^{2}R^{2}NUA is compared with six state-of-the-art nonlinear unmixing methods, including a BMM-based algorithm (i.e., PPNM [10]), four MLM-based algorithms (i.e., MLM [14], MLMp [15], R-MLM [31], and G-MLM [16]), and an EMLM-based algorithm (i.e., BNLSU [20]). PPNM [10] represents the PPNM-based unmixing algorithm considering only the second-order scatterings. As for MLM [14], MLMp [15], R-MLM [31], and G-MLM [16], they include all orders of interactions. Among them, MLM is the first method to perform unmixing based on the MLM; MLMp is an unsupervised nonlinear unmixing method, and we turn it into a supervised one for comparison, where the abundances and nonlinear parameters are estimated with the endmembers fixed; R-MLM adopts the $\ell_{2,1}$-norm-based loss function to enhance the robustness to noise; G-MLM imposes the graph constraints on the abundances and nonlinear parameters to exploit the internal manifold structure of the data and is run in a superpixel-by-superpixel manner. BNLSU [20] describes the nonlinear contributions at different bands and employs a graph constraint to make the band-neighboring nonlinear parameters similar. All algorithms are executed using MATLAB R2020b on a computer with Intel Core i9-9900K CPU at 3.60 GHz and 32-GB RAM.

The unmixing accuracy is evaluated using the root-mean-square error (RMSE) and the RE

$$\mathrm{RMSE} = \sqrt{\frac{1}{MN}\sum_{j=1}^{N}\left\|\boldsymbol{a}_j - \boldsymbol{a}_j\right\|_2^2} \quad (42)$$

$$\mathrm{RE} = \sqrt{\frac{1}{LN}\sum_{j=1}^{N}\left\|\boldsymbol{x}_j - \boldsymbol{x}_j\right\|_2^2} \quad (43)$$

where $\boldsymbol{a}_j$ and $\boldsymbol{x}_j$ represent the estimated abundance vector and spectrum for the jth pixel of HSIs, respectively. Since RMSE measures the proximity between the estimated and the ground truth, the smaller its value, the better the unmixing accuracy is. Since the ground truths of the real datasets are difficult to obtain, we use RE to help evaluate the unmixing accuracy. Note that a small RE only indicates the good data reconstruction ability but is not equivalent to yielding a good unmixing accuracy.



(a) (b)

Fig. 2. Simulated dataset: (a) five endmembers randomly selected from the USGS library and (b) true abundance maps.

In addition, we also compare the nonlinear intensity estimated by different algorithms. Specifically, the estimated nonlinear intensity can be roughly defined as the data reconstruction minus the estimated linear component, i.e., $\boldsymbol{X}_{\boldsymbol{Non}} = \boldsymbol{X} - \boldsymbol{X}_{\boldsymbol{Lin}} \in \mathbb{R}^{L\times N}$ [20]. Since the accuracy of nonlinear intensity estimation is not the ultimate goal of the unmixing task, we do not perform a numerical comparison, but visually compare the estimated nonlinear intensity from the spatial and spectral dimensions with two vectors, i.e., $\boldsymbol{NI}_{\mathrm{p}} \in \mathbb{R}^N$ and $\boldsymbol{NI}_{\mathrm{b}} \in \mathbb{R}^L$ defined as

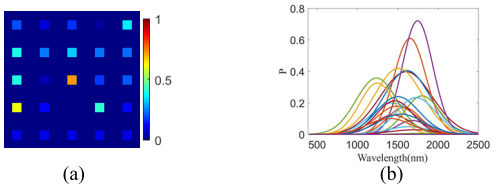
$$\begin{cases} \boldsymbol{NI}_{\mathrm{p}} = \sqrt{\dfrac{1}{L}\displaystyle\sum_{l=1}^{L}\left\|(\boldsymbol{X}_{\boldsymbol{Non}})_{l,:}\right\|_2^2} \\ \boldsymbol{NI}_{\mathrm{b}} = \sqrt{\dfrac{1}{N}\displaystyle\sum_{j=1}^{N}\left\|(\boldsymbol{X}_{\boldsymbol{Non}})_{:,j}\right\|_2^2}. \end{cases} \quad (44)$$

Then, we fold the vector $\boldsymbol{NI}_{\mathrm{p}}$ into a 2-D matrix according to spatial coordinates to generate the nonlinear intensity map. The curve of $\boldsymbol{NI}_{\mathrm{b}}$ is plotted to show the nonlinear intensity at different bands. Since the true nonlinear intensity is only available in the experiments on the simulated datasets, we compare the nonlinear intensity from the spatial and spectral dimensions. However, we do not compute the two vectors in the experiments on the semisimulated datasets, where the pixels show weak spatial correlation. In real scenes, the spatial distribution of the nonlinear intensity obeys a certain law, that is, strong nonlinear mixing effects often appear at the intersection of materials. Therefore, although the true nonlinear intensity is unavailable, we can compare the nonlinear intensity from the spatial dimension for experiments on the real datasets.

B. Experiments on Simulated Datasets

1) Data Generation: The DC consists of 75 × 75 pixels generated using five endmembers with 224 bands randomly selected from the U.S. Geological Survey (USGS) digital spectral library [56] shown in Fig. 2(a), and true abundances maps are given in Fig. 2(b). There are 25 squares arranged in the form of 5 × 5, and it is assumed that each square owns the homogeneous surface suggesting that the pixels in there have the same abundances. In the first row, each of the five

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(a) (b)

Fig. 3. Construction of the nonlinear parameters $\boldsymbol{P}$: (a) spatial distribution of $\boldsymbol{P}_{\max}$ and (b) single Gaussian distribution of $\boldsymbol{P}_j (j = 1, 2, \ldots, N)$ at each band.

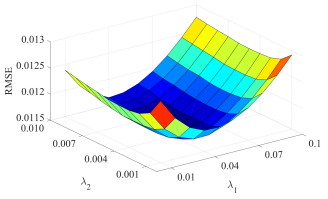


Fig. 4. RMSEs obtained by S^{2}R^{2}NUA with different parameter settings on DC1 with $\overline{\text{SNR}} = 30$ dB.

squares consists of one endmember. In the second row, two endmembers are mixed in the same proportion to generate five squares. As so on, five endmembers participating equally in the construction of the last five squares indicate the identity of these squares. In addition, the background pixels are mixtures of five endmembers with fixed abundances [36].

In the experiments, we generate DC based on the EMLM with the assumption that the distribution of $\boldsymbol{P}_j, j = 1, 2, \ldots, N$ is akin to the shape of the single Gaussian function [20]. Specifically, the procedures of constructing the vectorized nonlinear parameter matrix $\boldsymbol{P}$ are as follows: 1) values of vectorized nonlinear parameters belonging to the background pixels are set to zero, which implies that only the linear mixture exists; 2) assign the pixels of each squares the same maximum value $P_{\max,j}$ randomly selected from a half-normal distribution with $\sigma = 0.3$ [14] as shown in Fig. 3(a) (note that the last five identical squares have the same $P_{\max,j}$ [16]); and 3) the band index $\text{band}_{\max,j}$ of $P_{\max,j}$ is randomly selected from 70 to 150 [20], and then, $P_{l,j}$ at other bands can be accordingly calculated as follows:

$$P_{l,j} = P_{\max,j} \exp\left(-\frac{\left(l - \text{band}_{\max,j}\right)^2}{2\sigma_j^2}\right) \tag{45}$$

where σ_j is uniformly drawn from (20, 35), which regulates the amplitude of the vector $\boldsymbol{P}_j$'s distribution. The constructed $\boldsymbol{P}$ is shown in Fig. 3(b), which characterizes the smooth and rolling change of $P_{l,j}$ along with wavelengths.

To evaluate the robustness of the algorithms, the Gaussian noise is added to the simulated dataset, where the noise levels, represented by SNR, vary with bands to simulate the cases of real HSIs. A normal distribution $\text{SNR} \sim \mathcal{N}(\overline{\text{SNR}}, \epsilon^2)$ is used to assign the SNR for each band, where $\overline{\text{SNR}}$ is the average SNR of bands and $\epsilon = 5$ [25]. The first dataset, namely, DC1, is constructed with $\overline{\text{SNR}} = 20$, 30, and 40 dB. In addition, to examine the unmixing performance of the proposed method in the presence of highly noisy bands, we use two normal distributions $\text{SNR} \sim \mathcal{N}(\overline{\text{SNR}}_1, \epsilon^2)$ and $\text{SNR} \sim \mathcal{N}(\overline{\text{SNR}}_2, \epsilon^2)$ to construct another dataset, called DC2. While most bands have a common noise level with $\overline{\text{SNR}}_1 = 30$ dB, some randomly selected bands suffer from high-level average noise with $\overline{\text{SNR}}_2 = 15$ dB, and the number of noisy bands (NNBs) varies at 20, 30, and 40.

2) Selection of Parameters: To guarantee that the convergent and stable unmixing results can be obtained across different datasets, the stopping conditions for all the compared algorithms are set accordingly. When the multiple stopping conditions exist, the algorithm only needs to meet one of them to terminate. The PPNM is solved with the gradient decent (GD), in which the maximum number of iterations is 3000 and the tolerance is 10^{-8} [9]. The MLM is solved with the sequential quadratic programming (SQP), in which the function tolerance is 10^{-10} and the constraint tolerance is 10^{-8} [14]. The supervised MLMp is solved with a block coordinate descent (BCD) strategy, and the tolerance in MLMp is 10^{-8} [15]. The same stopping conditions, i.e., the maximum number of iterations is 1000 and the tolerance is 10^{-8}, are adopted for the three methods optimized via ADMM (i.e., R-MLM, G-MLM, and BNLSU). In R-MLM, the regularizer parameter of the abundances is 0.01 [31]. For G-MLM, the two regularizer parameters of the abundances are 0.001 and 0.05, respectively, and the regularizer parameter of the nonlinear parameters is 0.025 [16]. With regard to BNLSU, the regularizer parameters of the abundances and the vectorized nonlinear parameters are 0.001 and 0.3, respectively [20].

For the proposed algorithm, the outer and inner loops are set to 200 and 5, and the tolerance is 10^{-8}. Taking DC1 with $\overline{\text{SNR}} = 30$ dB as a studying case, we briefly analyze the impact of the two regularizer parameters in our algorithm before proceeding to the following experiments. The parameters $[\lambda_1, \lambda_2]$ are selected from ranges (0.01, 0.1) and (0.001, 0.01), respectively. In Fig. 4, the RMSEs obtained by S^{2}R^{2}NUA under different parameter configurations are depicted. Through observation, the change of RMSEs is gentle and we set $[\lambda_1, \lambda_2] = [0.03, 0.005]$. Note that our algorithm keeps the parameter settings in the following experiments.

In the SLIC method, there are two manually tuned parameters, namely, κ_s and κ, where κ_s weights the relative importance between the spectral and spatial similarities, and κ controls the average size of superpixels. As recommended in [41], we set κ_s to 0.3, provided that the spectral proximity contributes more to measuring similarity. Therefore, we only need to determine one parameter, i.e., κ.

To meet the over-segmentation needs, the value range of κ is set to (4, 10). The changes in the RMSEs yielded by our method concerning different κ are shown in Fig. 5(a). It can be seen that our method achieves the best RMSE when κ is 5. Furthermore, the unmixing accuracy is insensitive to changes in the value of κ over a relatively wide range. The generated superpixels regarding different κ are shown in Fig. 5(b)–(d). The superpixels that cover small squares contain fewer background pixels when κ is 5, which suggests that such a parameter setting is more appropriate. Therefore, we keep this parameter fixed throughout the experiments.

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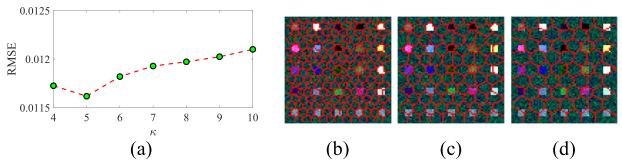


Fig. 5. Impact of κ on the unmixing accuracy: (a) RMSE obtained by S^{2}R^{2}NUA regarding different κ. Superpixel segmentation results on DC1 with $\overline{\text{SNR}} = 30$ dB regarding: (b) $\kappa = 5$; (c) $\kappa = 7$; and (d) $\kappa = 9$, where the RGB color images are composited by bands 56, 112, and 168.

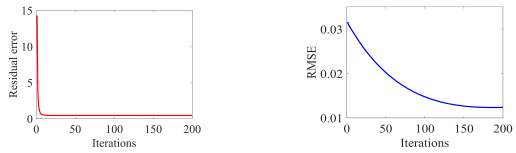


Fig. 6. Convergence analysis of S^{2}R^{2}NUA, by plotting the (Left) residual error and (Right) RMSE over 200 iterations.

3) Convergence Analysis: The ADMM with an outer–inner looping scheme is applied to optimize the unmixing problem, which is a popular method and has a good performance [37], [39]. The detailed proof of the convergence of ADMM can be easily found in [46]. It can be guaranteed that the proposed method will converge to a satisfactory solution after a certain number of iterations.

The changes of the residuals and RMSEs with the number of iterations are plotted in Fig. 6. After about 50 iterations, the residuals converge to small values, but the algorithm does not stop. By tracking the obtained RMSE, we can see that as the number of iterations increases, its value decreases more and more slowly, and gradually approaches the optimal value. In a word, the algorithm has a good convergence and can achieve optimal unmixing accuracy after about 200 iterations.

4) Ablation Experiments: To explicitly analyze the impacts of the $\ell_{2,1}$-norm-based loss function and the reweighted regularizers on the unmixing accuracy, we conduct the following ablation experiments.

First, to investigate the contributions of the two regularizers to the unmixing accuracy, we consider the following cases with different regularizers: 1) REMLM, an EMLM-based robust unmixing method using the $\ell_{2,1}$-norm-based loss function; 2) REMLM-W, REMLM with the regularizer on the abundances; and 3) S^{2}R^{2}NUA, REMLM with the two regularizers (the proposed method). All the above problems are optimized via ADMM. In REMLM, the maximum number of iterations is 1000 and the tolerance is 10^{-8}. The parameter settings of REMLM-W keep consistent with those of S^{2}R^{2}NUA, except for one regularizer parameter set to 0.03 for optimality, as shown in Fig. 7(a).

We run the above methods 20 times on DC1 in the case of $\overline{\text{SNR}} = 30$ dB with randomly distributed noise. The unmixing performances of these methods are evaluated with the average RMSEs, REs, and running times with the standard deviations (stds). In all tables, the best scores are bolded and underlined, and the suboptimal ones are only bolded. As shown in Table I, regarding the RMSEs, our method achieves the best result, followed by REMLM-W and worst by REMLM. It can be concluded that the utilization of the regularizers plays a role in improving the unmixing accuracy, and the best RMSE can be achieved when both the abundances and the nonlinear parameters are constrained. In addition, REMLM and REMLM-W produce smaller REs than our method, which may be attributed to overfitting. And, our method consumes more time due to employing the two regularizers.

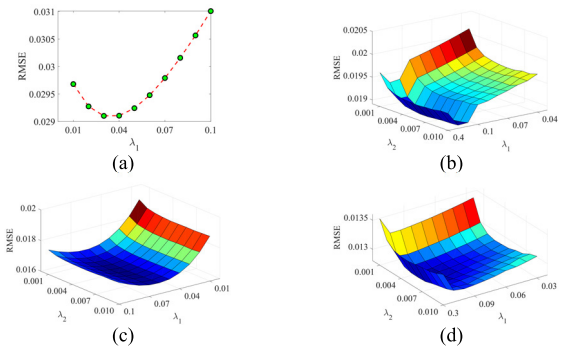


Fig. 7. Changes of the RMSEs with different parameters and the optimal settings for different methods: (a) REMLM-W, [0.03]; (b) W-EMLM, [0.2, 0.007]; (c) W^2-EMLM, [0.06, 0.006]; and (d) W-REMLM, [0.1, 0.005].

TABLE I

RMSEs, REs, and Running Times (Mean ± Std) Obtained by Methods With Different Regularizer Numbers Over 20 Runs on DC1 With $\overline{\text{SNR}} = 30$ dB

Metrics	REMLM	REMLM-W	S^{2}R^{2}NUA
RMSE	0.0295±0.0003	**0.0274±0.0003**	**0.0118±0.0011**
RE	**0.0002±0.0000**	**0.0002±0.0001**	0.0126±0.0007
Time (s)	**125.53±0.38**	**132.11±0.81**	232.78±0.32

TABLE II

Methods for Possible Combinations, Depending on Whether the $\ell_{2,1}$-Norm-Based Loss Function and the Reweighted Regularizers Are Employed or Not

Loss function		l_2-norm	$l_{2,1}$-norm
Without regularizers		EMLM	REMLM
With regularizers	Without reweighted factors	W-EMLM	W-REMLM
	With reweighted factors	W^2-EMLM	S^{2}R^{2}NUA (Proposed)

Further, we investigate the impacts of the $\ell_{2,1}$-norm-based loss function and the reweighting strategy on the unmixing performance by conducting more ablation experiments. In Table II, the possible combinations are listed, and the related methods are solved with ADMM. The maximum number of iterations and tolerance are set to 1000 and 10^{-8}, respectively, in the methods without using the reweighting strategy. Except for the regularizer parameters, the other parameters keep the same for these reweighted methods. As shown in Fig. 7(b)–(d), the RMSEs yielded by different methods under different parameter configurations are depicted, and their optimal parameter settings are given accordingly.

We compare the unmixing performance of the methods listed in Table II after running them 20 times on DC1 in the case of $\overline{\text{SNR}} = 30$ dB with randomly distributed noise. As shown in Table III, regarding the RMSEs, the results obtained by the three methods with the $\ell_{2,1}$-norm-based loss

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TABLE III

RMSEs, REs, AND RUNNING TIMES (MEAN ± STD) OBTAINED BY METHODS WITH DIFFERENT LOSS FUNCTIONS AND REGULARIZERS OVER 20 RUNS ON DC1 WITH $\overline{\text{SNR}} = 30$ dB

Metrics	EMLM	W-EMLM	W^2-EMLM	REMLM	W-REMLM	S^{2}R^{2}NUA
RMSE	0.0313 ±0.0002	0.0171 ±0.0015	0.0162 ±0.0013	0.0295 ±0.0003	**0.0130 ±0.0010**	**0.0118 ±0.0011**
RE	**0.0001 ±0.0000**	0.0127 ±0.0006	0.0124 ±0.0006	**0.0002 ±0.0000**	0.0129 ±0.0007	0.0126 ±0.0007
Time (s)	**96.20 ±0.29**	185.43 ±0.33	204.42 ±0.43	**125.53 ±0.38**	214.83 ±0.49	232.78 ±0.32

function are better than their counterparts, that is, the three methods with the ℓ_2-norm-based loss function. In addition, regardless of which loss function is based on, the methods with the reweighted regularizers achieve better results than the others. Overall, our method achieves the best result. It can be concluded that the unmixing accuracy can be improved to some extent by applying the $\ell_{2,1}$-norm-based loss function and the reweighted regularizers separately, while the best results can be obtained by applying them together. Prone to overfitting, EMLM and REMLM produce smaller REs than the other methods. Notably, our method benefits from the reweighting strategy, the spatial information of HSIs can be integrated into the nonlinear unmixing framework without taking too much time.

5) Robustness to Different Noise Levels: To investigate the robustness to different noise levels, all the compared algorithms are run 20 times on DC1 with randomly distributed noise. It can be seen from Table IV that the unmixing accuracy of different algorithms deteriorates with the decrease of $\overline{\text{SNR}}$. Among them, PPNM produces less competitive unmixing results than the other methods that consider higher order nonlinearity. Since MLM and MLMp take no account the prior information, they achieve limited unmixing accuracy and have weak robustness to noise. RMLM shows more stable and higher unmixing performance in most cases in comparison with other MLM-based methods. Running alone on each superpixel makes it difficult for G-MLM to obtain an optimal solution relative to the entire image, resulting in a certain decrease in the unmixing accuracy of G-MLM. BNLSU is prone to overfitting and is sensitive to noise. In all cases, it produces the best REs; however, the obtained RMSEs get dramatically worse along with $\overline{\text{SNR}}$ decreasing. Although our method is slightly worse than BNLSU in terms of the REs, it always yields the best RMSEs, indicating that our method achieves high-accuracy unmixing and is insensitive to noise. Notably, S^{2}R^{2}NUA, which exploits more prior information, consumes less time than BNLSU, indicating that our method can perform spectral–spatial unmixing more efficiently.

For illustration purposes, we take DC1 with an $\overline{\text{SNR}}$ of 30 dB as an example to further show the unmixing performance of different methods. In Fig. 8, we show the abundance maps generated by different algorithms. Due to the integration of the spatial information of HSIs, S^{2}R^{2}NUA generates spatially smooth abundance maps that are closest to the ground truth. While other methods can basically distinguish between endmembers, the loss or inaccurate of detailed information

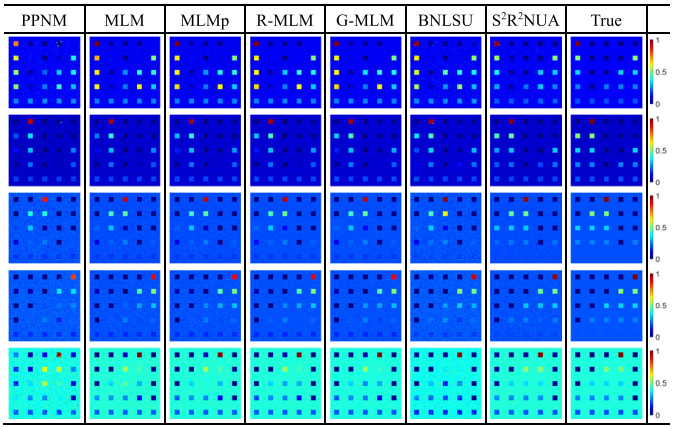


Fig. 8. Abundance maps generated by different methods on DC1 with $\overline{\text{SNR}} = 30$ dB.

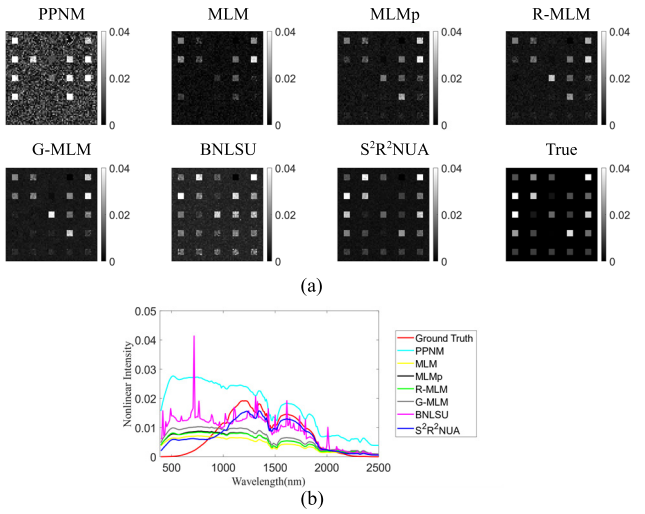


Fig. 9. Nonlinear intensity estimated by different methods on DC1 with $\overline{\text{SNR}} = 30$ dB: (a) nonlinear intensity at each pixel and (b) nonlinear intensity at each band.

can be observed from the corresponding abundance maps. In addition, we compare the estimated nonlinear intensity of different methods according to (44). Since S^{2}R^{2}NUA takes into account the local-region similarity of the vectorized nonlinear parameters, it obtains the most distinguishable map in Fig. 9(a), which can reflect the magnitude of the nonlinear intensity at different locations. The nonlinear intensity curves drawn by different methods are compared in Fig. 9(b). Ignoring that the nonlinear intensity varies across the bands, the unmixing methods based on the traditional spectral mixture models cannot correctly estimate the nonlinear contributions at different bands. Although BNLSU can model the wavelength-dependent nonlinear intensity, it is sensitive to noise, and its corresponding curve exhibits obvious glitches. On the contrary, S^{2}R^{2}NUA shows good robustness and obtains the curve closest to the ground truth. Further, we directly compare the nonlinear parameters obtained by BNLSU and S^{2}R^{2}NUA in Fig. 10. Our method inverts the nonlinear parameters accurately, while the result obtained by BNLSU deviates significantly from the ground truth.

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TABLE IV

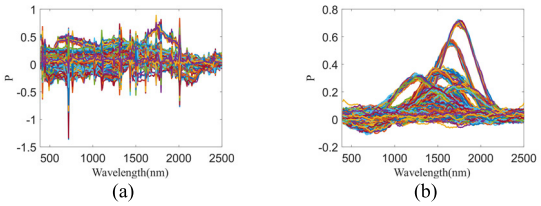
RMSEs, REs, AND RUNNING TIMES (MEAN ± STD) OBTAINED BY DIFFERENT METHODS OVER 20 RUNS ON DC1

SNR	Metrics	PPNM	MLM	MLMp	R-MLM	G-MLM	BNLSU	S^{2}R^{2}NUA
20 dB	RMSE	0.0591±0.0044	**0.0446±0.0018**	0.0597±0.0062	0.0505±0.0044	0.0681±0.0077	0.0574±0.0055	**0.0300±0.0021**
	RE	0.0430±0.0020	0.0431±0.0020	0.0416±0.0018	0.0419±0.0018	0.0416±0.0018	**0.0037±0.0003**	**0.0405±0.0019**
	Time (s)	64.88±1.03	**14.69±0.11**	**6.23±0.56**	128.75±1.40	40.28±3.47	271.72±6.87	232.44±4.21
30 dB	RMSE	0.0366±0.0011	0.0322±0.0002	0.0305±0.0002	**0.0296±0.0005**	0.0300±0.0002	0.0316±0.0008	**0.0118±0.0011**
	RE	0.0145±0.0006	0.0147±0.0006	0.0146±0.0006	0.0146±0.0006	0.0146±0.0006	**0.0063±0.0001**	**0.0126±0.0007**
	Time (s)	47.43±2.32	**13.96±0.11**	**3.26±0.06**	128.68±0.35	24.33±0.58	272.55±1.77	232.78±0.32
40 dB	RMSE	0.0333±0.0003	0.0308±0.0000	0.0292±0.0000	0.0275±0.0002	0.0294±0.0000	**0.0262±0.0002**	**0.0059±0.0004**
	RE	0.0069±0.0001	0.0073±0.0001	0.0072±0.0001	0.0073±0.0001	0.0072±0.0001	**0.0033±0.0001**	**0.0038±0.0002**
	Time (s)	32.08±1.63	**13.67±0.07**	**5.01±0.11**	128.55±0.26	22.31±0.35	272.58±0.34	232.78±0.33

TABLE V

RMSEs, REs, AND RUNNING TIMES (MEAN ± STD) OBTAINED BY DIFFERENT METHODS OVER 20 RUNS ON DC2

NNB	Metrics	PPNM	MLM	MLMp	R-MLM	G-MLM	BNLSU	S^{2}R^{2}NUA
20	RMSE	0.0424±0.0030	0.0359±0.0017	0.0360±0.0042	**0.0300±0.0009**	0.0364±0.0050	0.0428±0.0166	**0.0141±0.0012**
	RE	0.0257±0.0021	0.0258±0.0021	0.0254±0.0020	0.0258±0.0022	0.0254±0.0020	**0.0063±0.0007**	**0.0243±0.0022**
	Time (s)	56.08±3.26	**14.40±0.19**	**4.15±0.75**	129.07±1.45	29.49±2.26	274.60±2.46	235.68±2.77
30	RMSE	0.0476±0.0070	0.0374±0.0018	0.0396±0.0050	**0.0301±0.0007**	0.0416±0.0063	0.0525±0.0148	**0.0145±0.0010**
	RE	0.0305±0.0030	0.0306±0.0030	0.0301±0.0028	0.0306±0.0030	0.0301±0.0028	**0.0068±0.0013**	**0.0293±0.0031**
	Time (s)	59.81±4.69	**14.51±0.29**	**4.66±0.76**	128.42±0.89	32.31±3.32	273.10±0.80	233.94±0.46
40	RMSE	0.0504±0.0088	0.0397±0.0033	0.0481±0.0115	**0.0303±0.0008**	0.0523±0.0133	0.0558±0.0150	**0.0158±0.0011**
	RE	0.0358±0.0043	0.0359±0.0043	0.0350±0.0039	0.0358±0.0043	0.0349±0.0039	**0.0072±0.0021**	**0.0345±0.0043**
	Time (s)	61.13±3.36	**14.61±0.15**	**5.51±1.05**	128.59±0.98	36.78±4.42	273.13±0.57	235.03±0.80



(a) (b)

Fig. 10. Nonlinear parameters calculated by the two EMLM-based unmixing methods on DC1 with $\overline{\text{SNR}} = 30$ dB: (a) BNLSU and (b) S^{2}R^{2}NUA.

6) Robustness to the NNBs: To examine the unmixing performance of all the compared algorithms when there are a number of highly noisy bands in the data, we run these algorithms 20 times on DC2 with randomly distributed noise. As shown in Table V, the RMSEs and REs obtained by different algorithms are worse than those obtained on DC1 correspondingly, due to the fact that a number of bands are corrupted by stronger noise. It is not surprising that all the algorithms have a decline in the unmixing accuracy when the number of highly noisy bands increases. To be specific, the three unregularized methods (i.e., PPNM, MLM, and MLMp) cannot achieve high unmixing accuracy due to the lack of utilization of the prior information. R-MLM gets the suboptimal RMSEs since it can effectively suppress the effect of the noisy bands. G-MLM is sensitive to noise, resulting in its poor unmixing accuracy. In terms of the REs, BNLSU produces the best scores. However, BNLSU yields the unsatisfactory RMSEs, and it has unstable results when the noise intensity varies. It can be observed that S^{2}R^{2}NUA yields the optimal RMSEs and the suboptimal REs in all cases, which validates that our method can achieve good and stable unmixing results even when the hyperspectral data suffer from

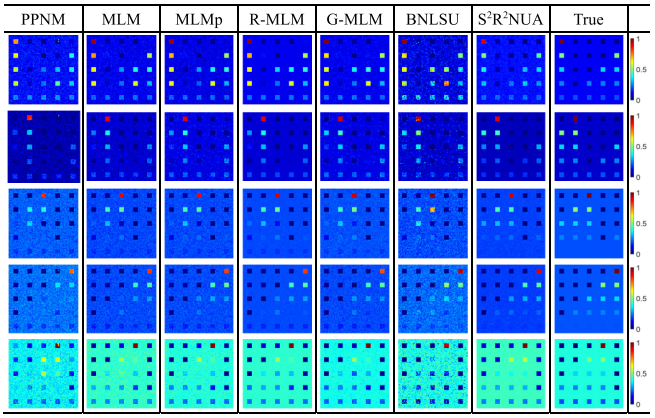


Fig. 11. Abundance maps generated by different methods on DC2 with NNB = 30.

severe noise. We show the running times of different methods and draw the similar conclusion to Section IV-B5 that our algorithm achieves a satisfactory unmixing performance in an acceptable time.

In addition, taking the DC2 containing 30 highly noise bands as an example, we show the abundance maps generated by different algorithms in Fig. 11. It can be obviously seen that S^{2}R^{2}NUA is less affected by noise and yields fine abundance maps, while the maps generated by the others are coarse. Likewise, in Fig. 12, the nonlinear intensities estimated by different algorithms are compared from the spatial and spectral dimensions, respectively. It can be observed that the nonlinear intensity map and curve generated by our method are closest to the ground truth. Fig. 13 shows the vectorized nonlinear parameters estimated by BNLSU and the proposed method. It can be observed that S^{2}R^{2}NUA can achieve accurate

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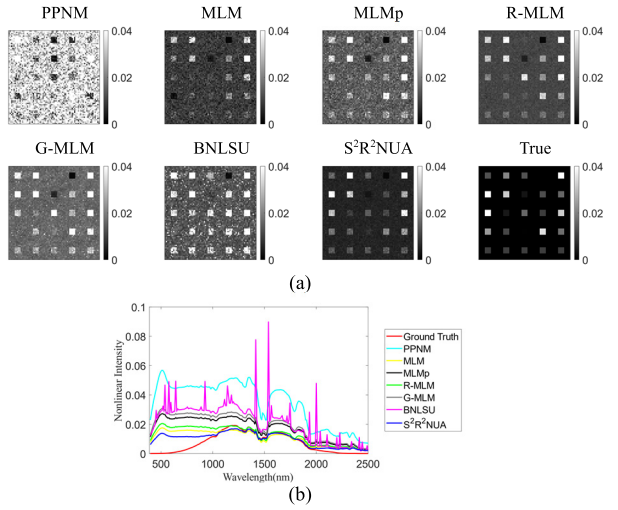


Fig. 12. Nonlinear intensity estimated by different methods on DC2 with NNB = 30: (a) nonlinear intensity at each pixel and (b) nonlinear intensity at each band.

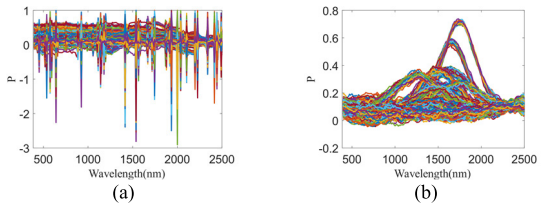


Fig. 13. Nonlinear parameters calculated by the two EMLM-based unmixing methods on DC2 with NNB = 30: (a) BNLSU and (b) S^2R^2NUA.

inversion of the vectorized nonlinear parameters even when the hyperspectral data suffer from severe noise. Conversely, the results of BNLSU are further deteriorated due to its sensitivity to noise. Overall, we can reach similar conclusions as above that the proposed method achieves high-accuracy unmixing and is insensitive to noise.

C. Experiments on Semisimulated Datasets

1) Data Description: Although the experiments on simulated datasets validated the unmixing performance of the proposed method, these DCs are generated based on the EMLM, which may not exactly match the real-world situation. Therefore, we conduct experiments on four semisimulated forest datasets, i.e., Frst-b20p80, Frst-b40p60, Frst-b60p40, and Frst-b80p20, consisting of two types of trees, i.e., beech trees and poplar trees, and a minor amount of soil. As shown in Fig. 14(a), the proportion of the two trees in each pixel varies from 20% to 80% for different datasets. Detailed descriptions of these datasets can be found in [20]. Taking into account the spectral variability, corresponding endmember matrices shown in Fig. 14(b) are applied to unmix each pixel. Since the semisimulated datasets show weak spatial correlations [57], we run G-MLM on the whole image, not in a superpixel-by-superpixel manner. Notably, to reasonably run the proposed method on these datasets, we remove the outer loop module for

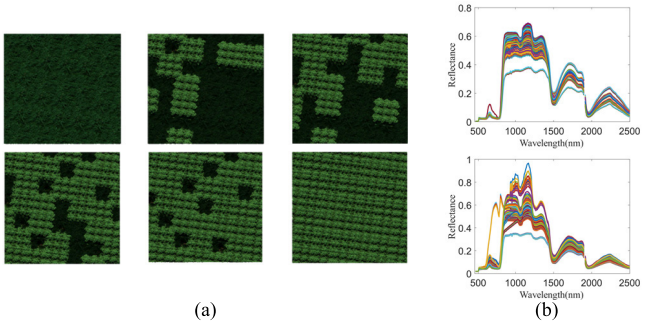


Fig. 14. Forest datasets: (a) (First row): 100% beech tree + 0% poplar tree, 80% beech tree + 20% poplar tree, and 60% beech tree + 40% poplar tree and (Second row): 40% beech tree + 60% poplar tree, 20% beech tree + 80% poplar tree, and 0% beech tree + 100% poplar tree. (b) Spectral curves in each pixel, (Top) beech and (Bottom) poplar.

TABLE VI
RMSEs AND REs OBTAINED BY DIFFERENT METHODS ON THE SEMISIMULATED DATASETS

Datasets	Metrics	PPNM	MLM	MLMp	R-MLM	G-MLM	BNLSU	S^2R^2NUA
Frst -b20p80	RMSE	0.0403	0.0470	0.0516	0.0397	**0.0372**	0.0385	**0.0311**
	RE	0.0025	0.0042	0.0045	0.0050	0.0048	**0.0043**	**0.0018**
Frst -b40p60	RMSE	0.0547	0.0603	0.0632	0.0541	0.0511	**0.0424**	**0.0399**
	RE	0.0025	0.0029	0.0029	0.0030	0.0036	**0.0017**	**0.0017**
Frst -b60p40	RMSE	0.0618	0.0630	0.0634	0.0598	0.0856	**0.0348**	**0.0342**
	RE	0.0021	0.0020	0.0021	0.0021	0.0036	**0.0007**	**0.0018**
Frst -b80p20	RMSE	0.0586	0.0627	0.0635	0.0576	0.0569	**0.0356**	**0.0259**
	RE	0.0021	0.0022	0.0022	0.0023	0.0034	**0.0007**	**0.0018**

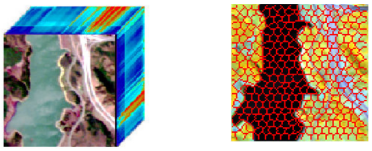


Fig. 15. Jasper Ridge: (a) DC and (b) generated superpixels, where the RGB image is composited by bands 53, 102, and 170.

updating the reweighting factors and set the maximum number of iterations of the inner loop module to 1000 while keeping other parameter settings unchanged.

2) Experimental Results and Analysis: In Table VI, since PPNM is suitable for dealing with the unmixing problem under the vegetation-soil scenario, it can obtain good results. In most cases, R-MLM and G-MLM outperform the two unregularized MLM-based unmixing methods (i.e., MLM and MLMp). Both BNLSU and S^2R^2NUA yield satisfactory unmixing results due to the consideration of wavelength-dependent nonlinear intensity, with S^2R^2NUA being better. In conclusion, the proposed method shows great superiority in unmixing the semisimulated HSIs. However, due to the lack of meaningful spatial information in these datasets, the experiments only verify the effectiveness of the inner loop module of the proposed method without using reweighted factors. Therefore, further experiments on real HSIs are required.

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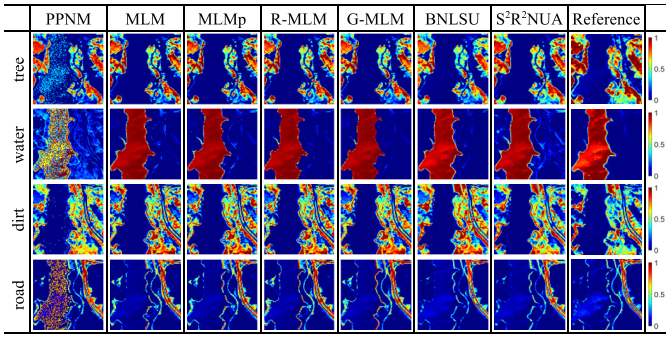


Fig. 16. Abundance maps of four endmembers generated by different methods on Jasper Ridge dataset.

D. Experiments on Real Datasets

1) Jasper Ridge: In remote sensing, Jasper Ridge dataset has been extensively investigated. In Fig. 15(a), we select a subimage with 100 × 100 pixels to evaluate the unmixing performance. Each pixel is recorded at 224 bands ranging from 380 to 2500 nm with spectral resolution up to 9.46 nm. After removing 1–3, 108–112, 154–166, and 220–224 bands contaminated by dense water vapor and atmospheric effects, 198 bands have finally remained for unmixing. There feature primarily four endmembers, i.e., tree, water, dirt, and road in the scene, and the VCA is applied to extract the endmembers. The generated superpixels are shown in Fig. 15(b).

We evaluate all unmixing methods with the RMSEs and REs, taking the ground truth of Jasper Ridge dataset provided in [58] as a reference. In Table VII, it can be seen that PPNM gets the worst scores. Similar unmixing performance can be yielded by MLM, MLMp, and R-MLM. Since we run G-MLM in a superpixel-by-superpixel manner, the discontinuity of the input data degrades the unmixing accuracy. Although BNLSU gets the best RE, it does not perform as well as our method in the RMSEs. Furthermore, we compare abundance maps generated by different algorithms in Fig. 16. S^2R^2NUA obtains fine abundance maps, in which the dirt endmember's map is more discriminative than that generated by other methods. From Fig. 17, it can be observed that the nonlinear intensity map produced by our method is spatially smooth and shows sharp boundaries, which coincides with the fact that the interactions between materials often occur in adjacent regions. Jasper Ridge dataset corresponds to a relatively simple scenario, and to further demonstrate the superiority of the proposed method, we investigate another more challenging real hyperspectral data.

2) Cuprite: As shown in Fig. 18(a), we survey a widely analyzed 250 × 190 subimage taken from the Cuprite mining image [25]. The recorded data contain 224 bands, covering a wavelength range of 0.4–2.5. After removing the noisy and water absorption bands, i.e., 1–3, 105–115, 150–170, and 223–224, 187 bands are finally remained. It is speculated that the geographic composition of the region includes up to 14 minerals. Neglecting similar signatures, we consider 12 endmembers. The VCA is used for endmember extraction. The generated superpixels are shown in Fig. 18(b).

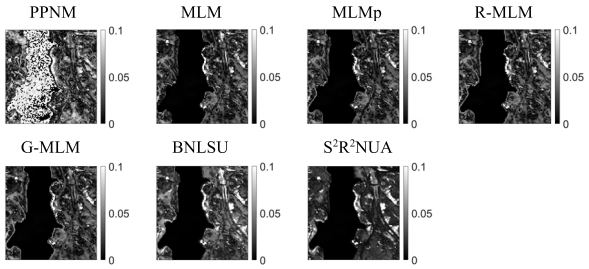


Fig. 17. Nonlinear intensity maps generated by different methods on Jasper Ridge dataset.

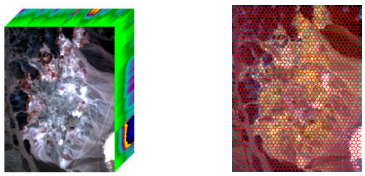


Fig. 18. Cuprite: (a) DC and (b) generated superpixels, where the RGB image is composited by bands 50, 97, and 175.

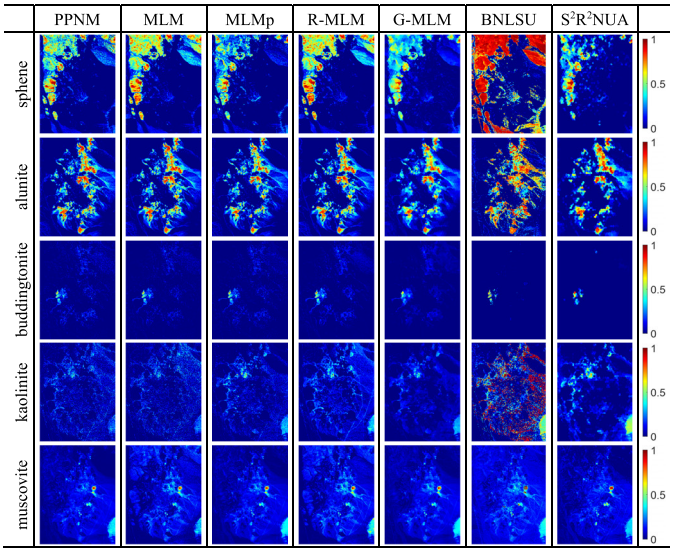


Fig. 19. Abundance maps of five endmembers generated by different methods on Cuprite dataset.

TABLE VII
RMSEs AND REs OBTAINED BY DIFFERENT METHODS ON JASPER RIDGE DATASET

Metrics	PPNM	MLM	MLMp	R-MLM	G-MLM	BNLSU	S^2R^2NUA
RMSE	0.2263	0.1308	0.1316	0.1317	0.1357	**0.1244**	**0.1097**
RE	0.0248	0.0209	0.0211	0.0213	0.0211	**0.0041**	**0.0081**

Since the ground truth remains unknown in the scene, we evaluate the unmixing method by means of comparing the REs obtained by different methods. It can be observed from Table VIII that the proposed method achieves the best RE. The abundance maps of the five representative endmembers (i.e., sphene, alunite, buddingtonite, kaolinite, and muscovite) generated by different methods are shown in Fig. 19.

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TABLE VIII
REs OBTAINED BY DIFFERENT METHODS ON CUPRITE DATASET

Metrics	PPNM	MLM	MLMp	R-MLM	G-MLM	BNLSU	S^2R^2NUA
RE	0.0015	**0.0014**	0.0015	0.0015	0.0017	0.0015	**0.0012**

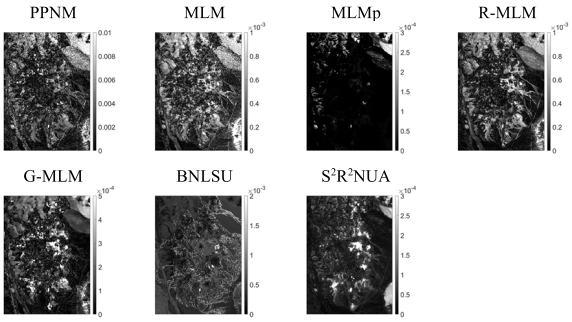


Fig. 20. Nonlinear intensity maps generated by different methods on Cuprite dataset.

Although most of the methods can distinguish these endmembers, their corresponding abundance maps are too coarse. Notably, the abundance maps generated by BNLSU are inaccurate due to incorrect unmixing, whereas the ones generated by S^2R^2NUA are more refined. It can be seen from Fig. 20 that the proposed method obtains a spatially smooth nonlinear intensity map, where concentrated nonlinear intensities can be observed at the material boundaries.

E. Discussion

According to the experimental results on the simulated, semisimulated, and real datasets, we can summarize the advantages of the proposed method as follows.

1) Higher Unmixing Accuracy: To handle the nonlinear unmixing problem in real scenarios, the proposed method pays attention to the inherent wavelength-dependent nonlinear intensity and noise effects. The nonlinear contributions at the pixels' different bands are modeled with the EMLM, while the noise effects are suppressed by using a robust metric (i.e., the $\ell_{2,1}$-norm) for the RE. Fully exploiting the rich spectral–spatial information of HSIs contributes to mitigating overfitting and improving unmixing accuracy. Since the similar surface covers often occur in the local regions, the reweighted collaborative sparse regularizer of the abundances and the reweighted spectral TV regularizer of the vectorized nonlinear parameters are well-designed, respectively, and then, integrated to the nonlinear unmixing problem. Consequently, the proposed method achieves higher unmixing accuracy than other state-of-the-art methods on the simulated, semisimulated, and real hyperspectral datasets. In addition, our method generates the spatially smooth abundance maps as well as a nonlinear intensity map, which coincides with the real-world situations.

2) Stronger Robustness to Noise: The experiments are conducted on the two simulated datasets, and the results demonstrate that the proposed method shows stronger robustness to different noise levels and different NNBs, in comparison with other state-of-the-art methods. Since our method can provide the accurate abundance estimation and nonlinear parameter inversion, even when the hyperspectral data suffer from severe noise, it should be very useful for practical applications.

3) Goodness of the Reweighting Strategy: Instead of imposing additional parameter-sensitive spatial regularizers, the proposed method uses the two regularizers to integrate such rich prior information into the nonlinear unmixing problem, which facilitates spectral–spatial unmixing in nonlinear mixing scenarios.

4) Convenience in Selecting Parameters: Although there are only two manually tuned regularizer parameters in S^2R^2NUA, they remain unchanged throughout the experiments. Satisfactory unmixing results have been achieved by the proposed S^2R^2NUA under this setting for all the employed hyperspectral datasets. The simplicity of the parameter setting also makes our method convenient for practical applications.

V. CONCLUSION

In this article, we have proposed an S^2R^2NUA. It aims to solve two major problems: how to enhance the robustness of the nonlinear unmixing algorithm to noise and how to effectively integrate the spectral–spatial information of HSIs into nonlinear unmixing. First, to model the wavelength-dependent nonlinear intensity and suppress the effects of noise, we have constructed an EMLM-based robust unmixing method where the data RE is measured with the $\ell_{2,1}$-norm. In addition, the similarities of the surface cover in the local regions are well-exploited. To this end, we have designed the reweighted collaborative sparse regularizer of the abundances and the reweighted spectral TV regularizer of the vectorized nonlinear parameters, and then incorporated them into the nonlinear unmixing problem. The two regularizers express the physical realities of HSIs that pixels within a superpixel-based neighborhood share the same subset of endmembers with similar abundances and the vectorized nonlinear parameters have spectral–spatial smoothness, respectively. Finally, ADMM is applied to solve the constrained optimization problem. The experimental results on the simulated, semisimulated, and real hyperspectral datasets have shown that the proposed method can achieve higher unmixing accuracy and exhibits stronger robustness to noise compared to the other state-of-the-art unmixing methods.

The proposed method is currently inaccessible for endmember extraction. However, the joint estimation of endmembers and abundances is usually required in practical applications. Therefore, in future studies, we will focus on improving the proposed method to be suitable for unsupervised nonlinear unmixing tasks. In addition, how to effectively incorporate the nonlocal spatial similarity of HSIs into the nonlinear unmixing problem to further improve the unmixing accuracy may be another topic worthy of studying.

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